Vision models on Microcontrollers

Marco Zennaro, PhD
The Abdus Salam International Centre for Theoretical Physics –Italy

Credit

- All the credit goes to:

  Brian Plancher, Dartmouth College,Accessible and Accelerated Robotics Lab (A^2R Lab)

- https://brianplancher.com

- Original video:

  https://www.youtube.com/watch?v=Grf_21CXgo4

Computer Vision is Hard

What color are the pants and the shirt?



Slide Credit: Hamilton Chong

Computer Vision is Hard



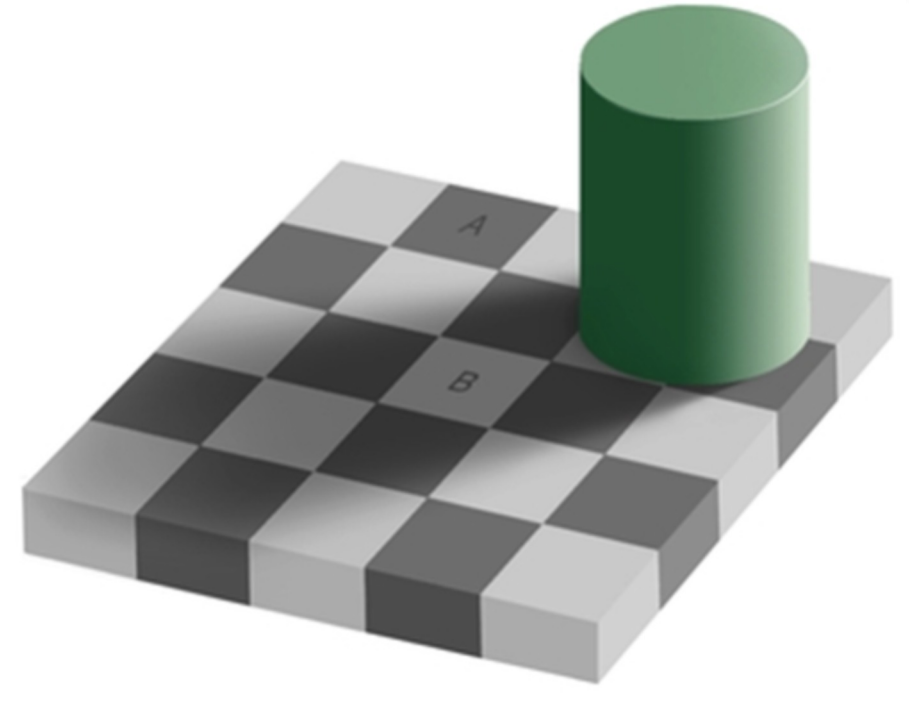
Slide Credit: Hamilton Chong

Computer Vision is Hard



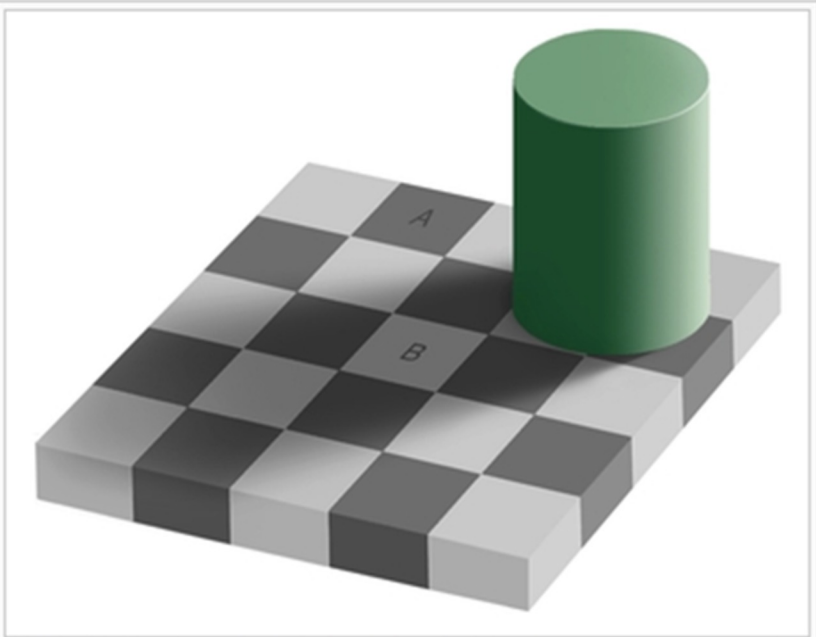
Slide Credit: Hamilton Chong

Computer Vision is Hard

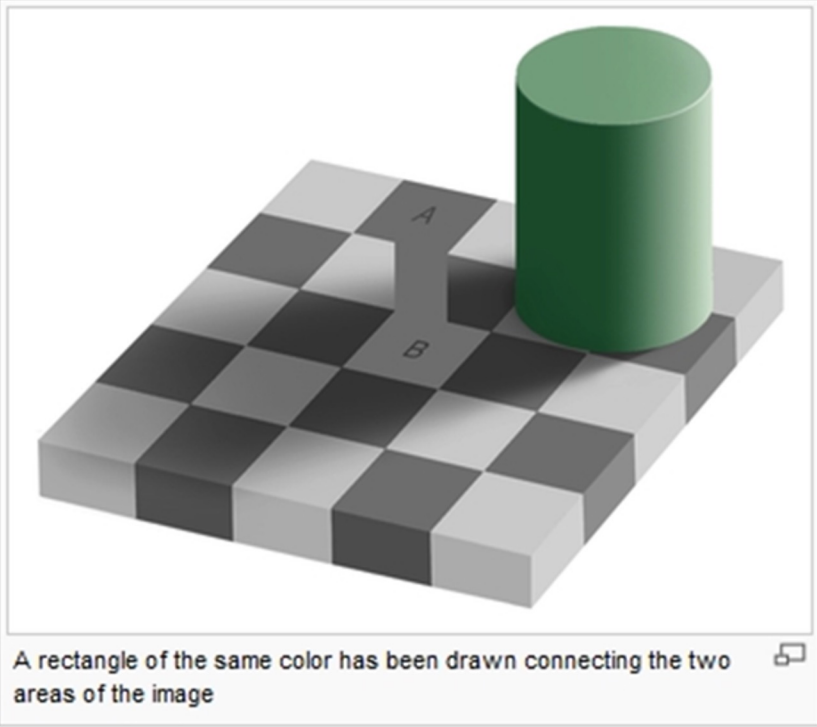


Is square A or B darker in color?

Computer Vision is Hard

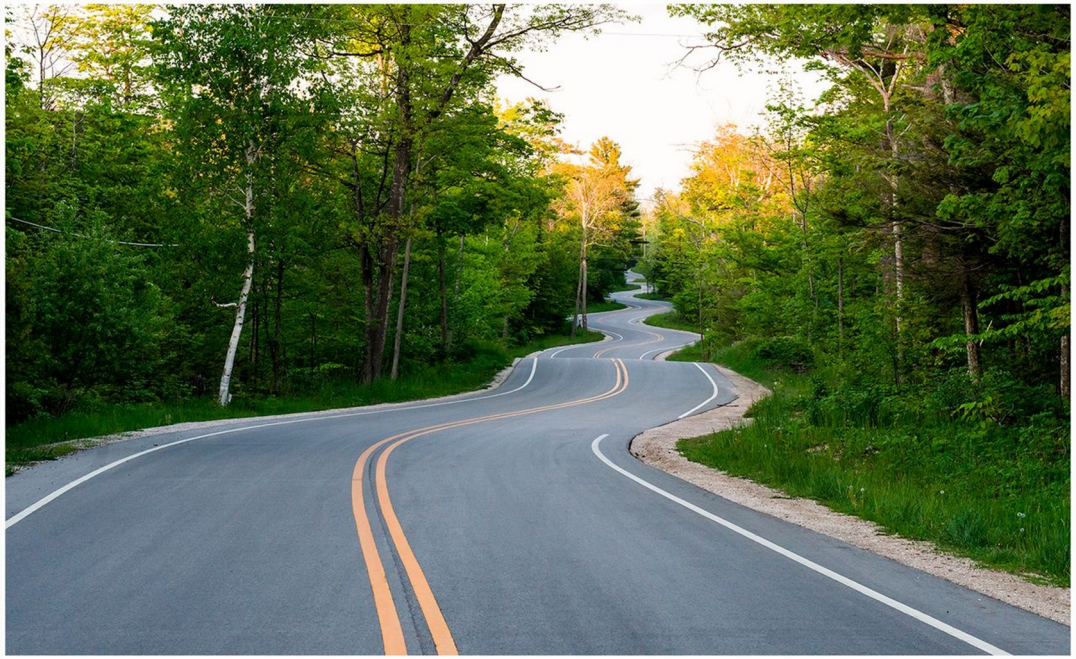


Areas of the image A and B are the same color



A rectangle of the same color has been drawn connecting the two areas of the image

What Features of the image might be important for self driving cars?



What Features of the image might be important for self driving cars?

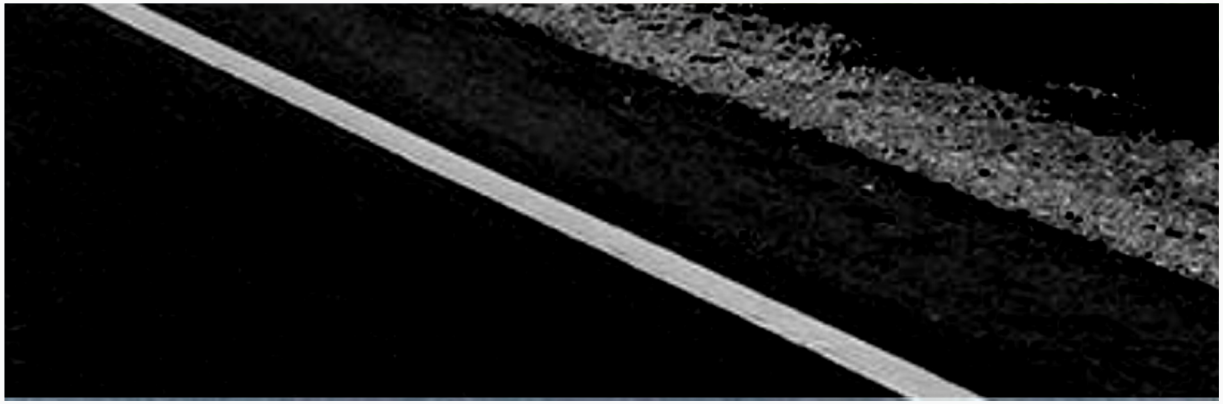


Maybe straight lines to see the lanes of the road?

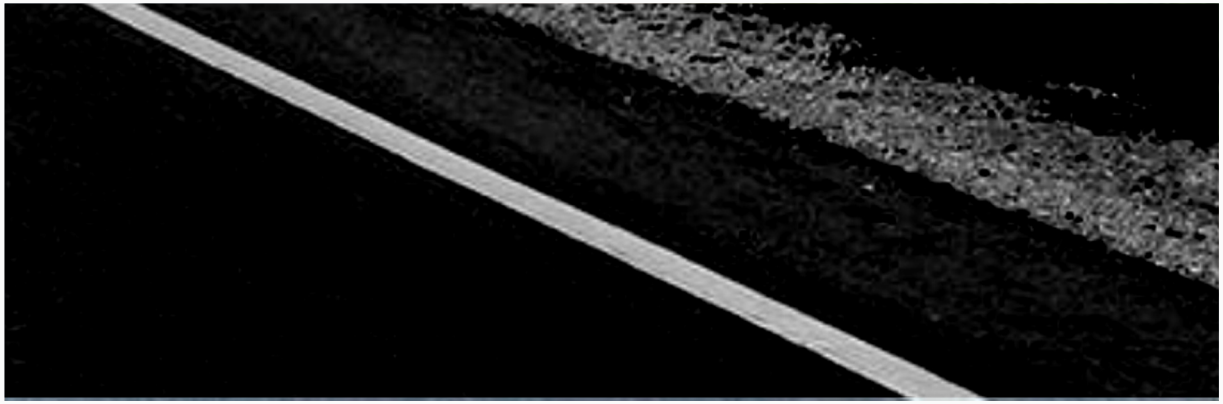
How might we find these features?



How might we find these features?

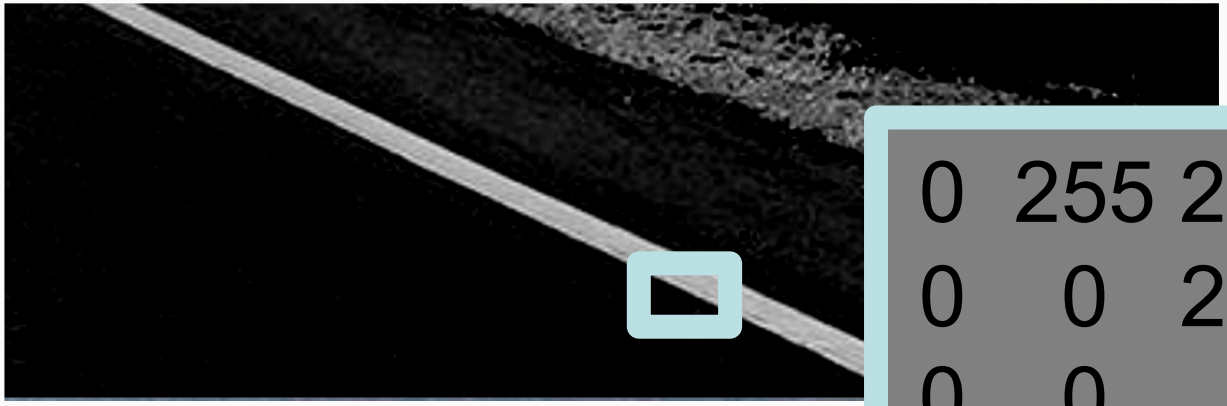


How might we find these features?



Black: 0
White: 255

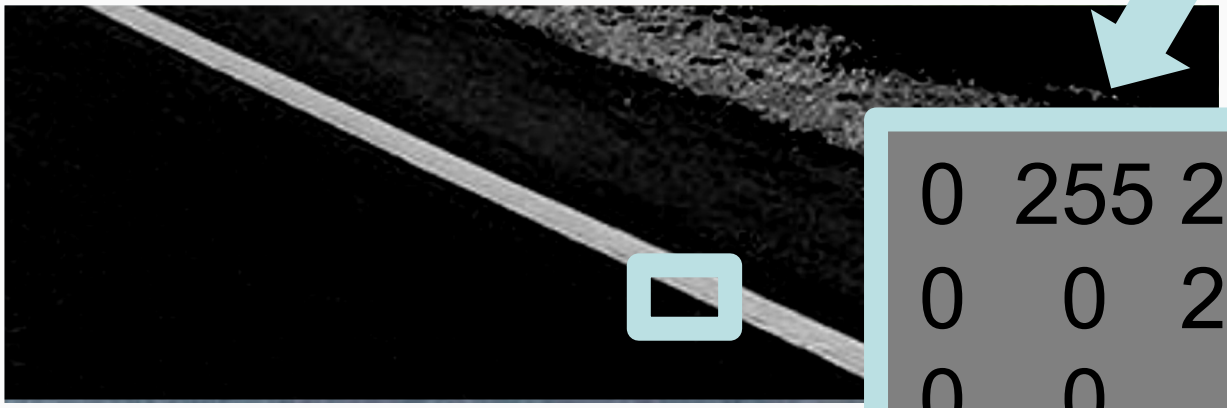
How might we find these features?



0	255	255	255	255
0	0	255	200	255
0	0	0	255	255
0	0	0	0	255
0	0	0	0	0

Black: 0
White: 255

How might we find these features?



Look for a Big Change!

0	255	255	255	255
0	0	255	200	255
0	0	0	255	255
0	0	0	0	255
0	0	0	0	0

Black: 0
White: 255

How might we find these features? Convolutions

How might we find these features? Convolutions

Original Image

0	0	0	255	255	255
0	0	0	255	255	255
0	0	0	255	255	255
0	0	0	255	255	255
0	0	0	255	255	255
0	0	0	255	255	255

How might we find these features? Convolutions

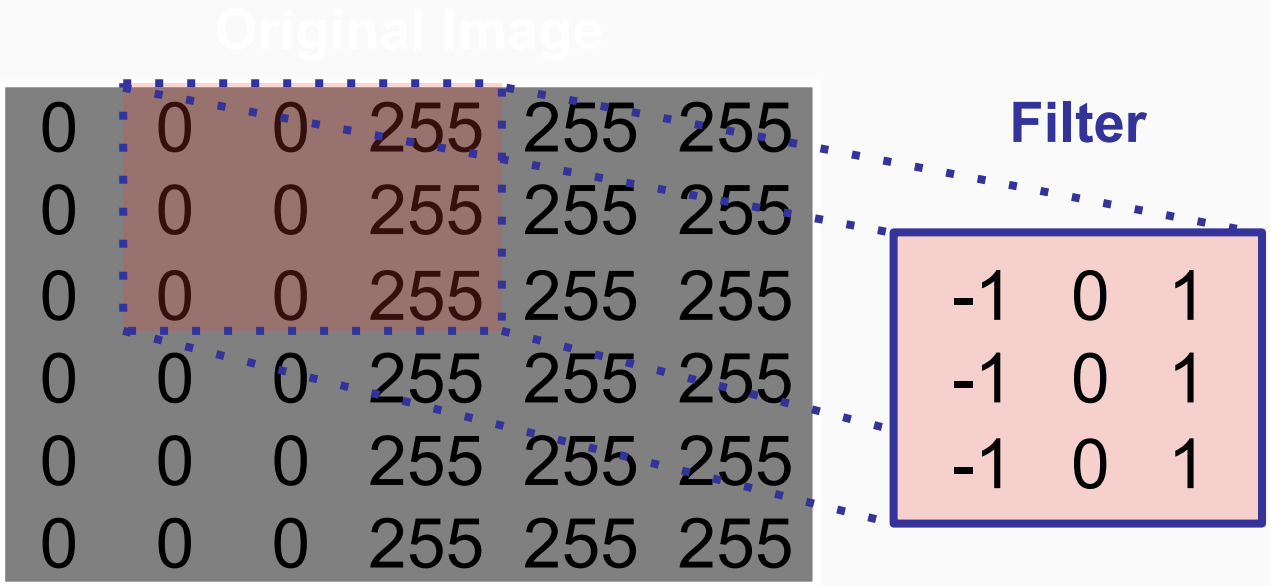
Original Image

0	0	0	255	255	255
0	0	0	255	255	255
0	0	0	255	255	255
0	0	0	255	255	255
0	0	0	255	255	255
0	0	0	255	255	255

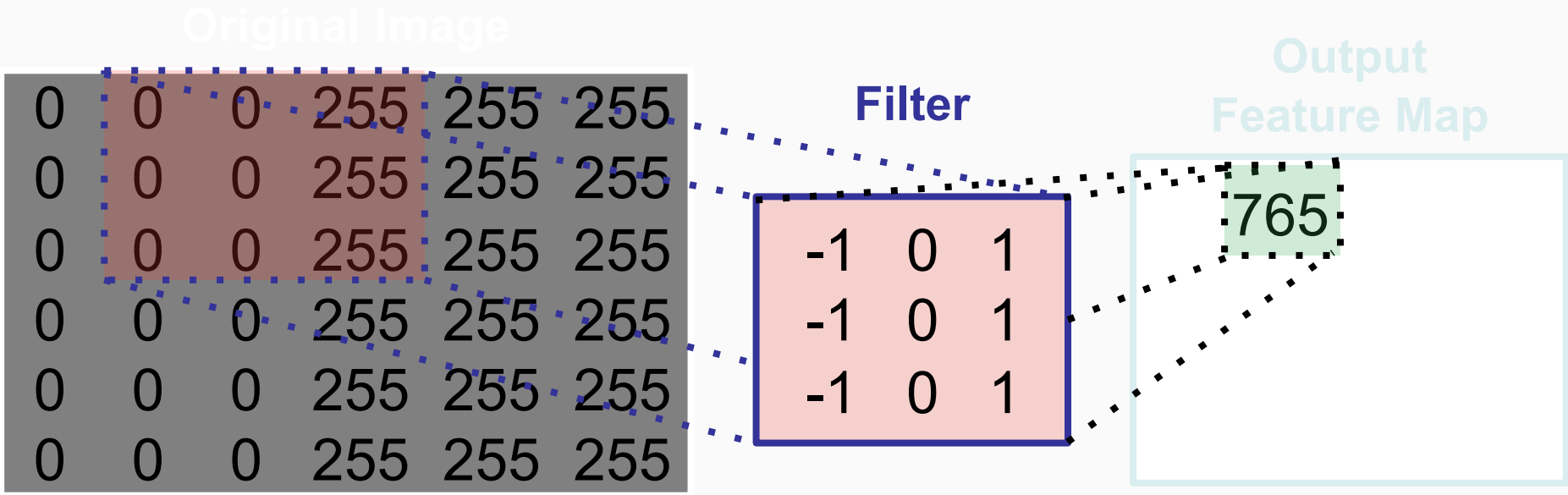
Filter

-1	0	1
-1	0	1
-1	0	1

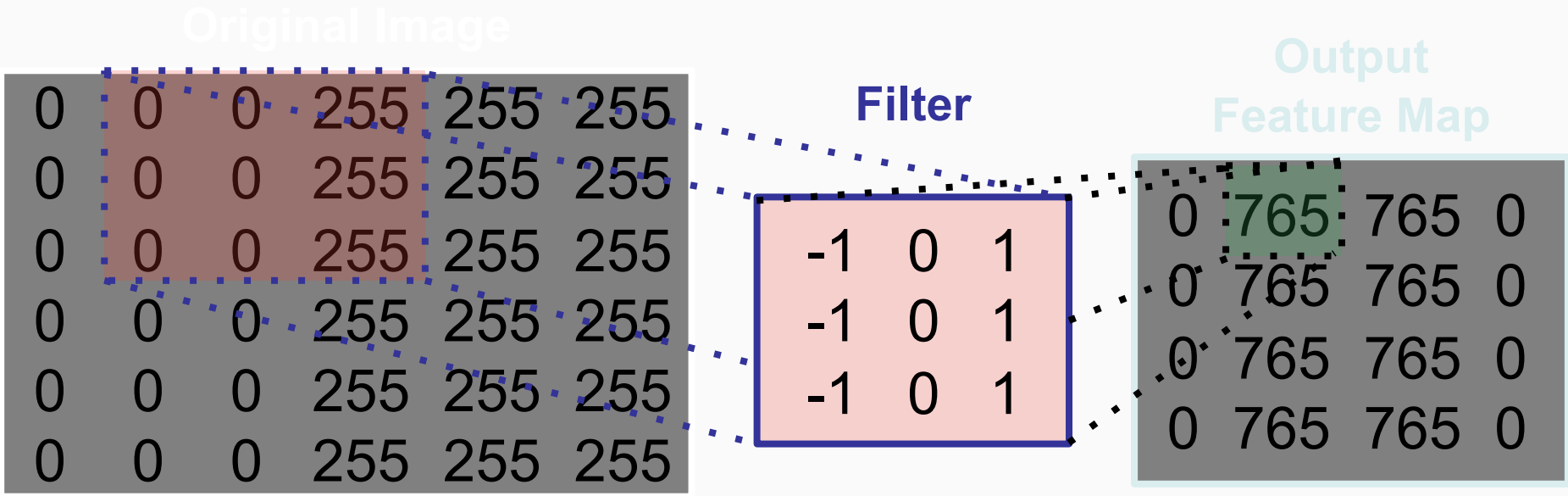
How might we find these features? Convolutions



How might we find these features? Convolutions



How might we find these features? Convolutions



How might we find these features? Convolutions

Colab Link



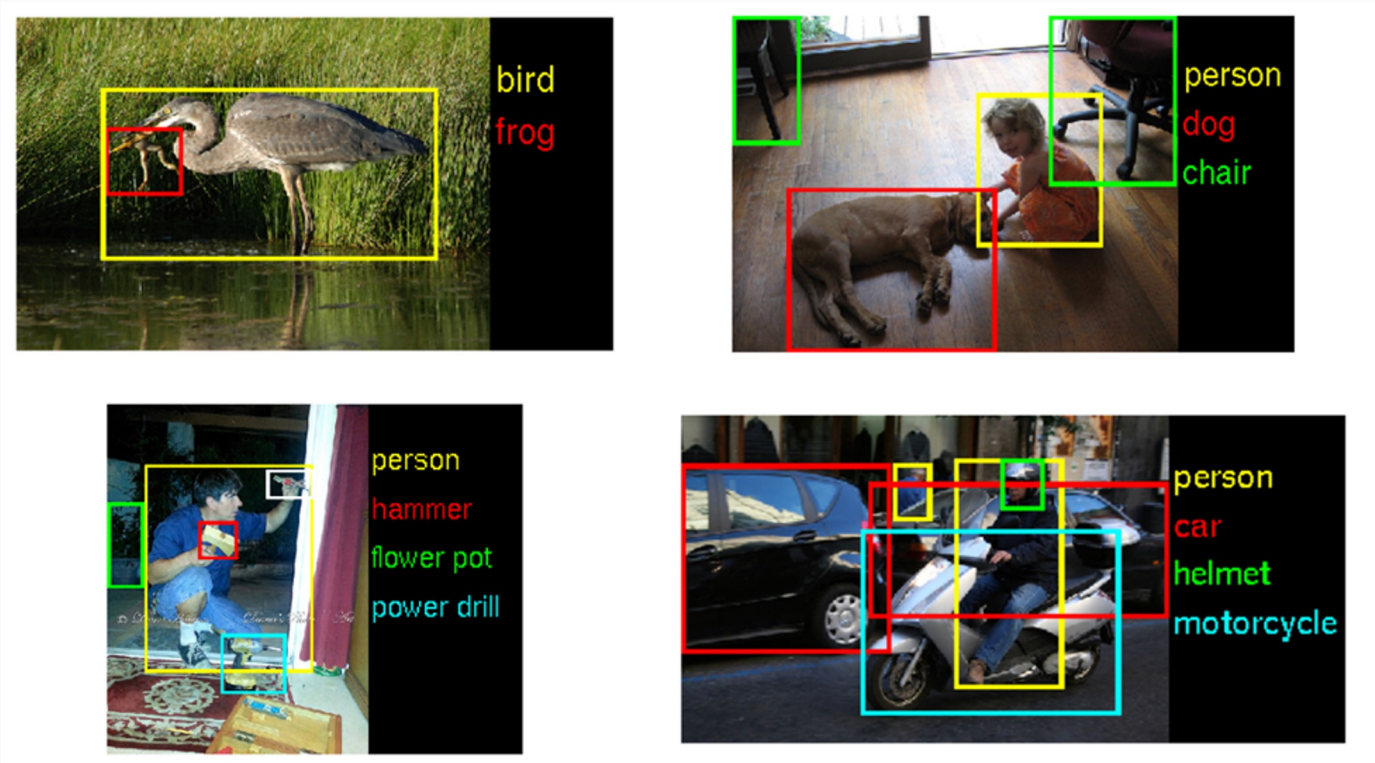
-1	0	1
-2	0	2
-1	0	1

Features

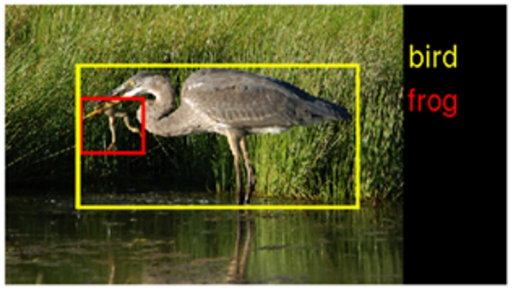


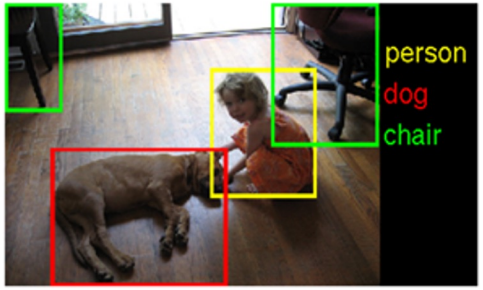
https://setosa.io/ev/image-kernels/

What features are needed for Object Detection?

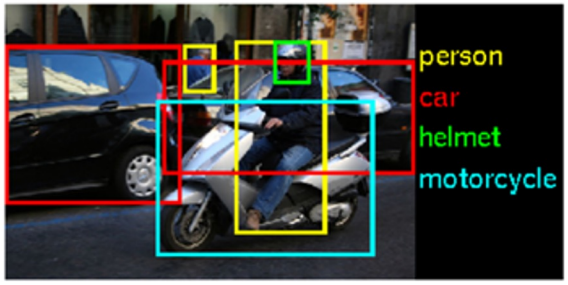


What features are needed for Object Detection?



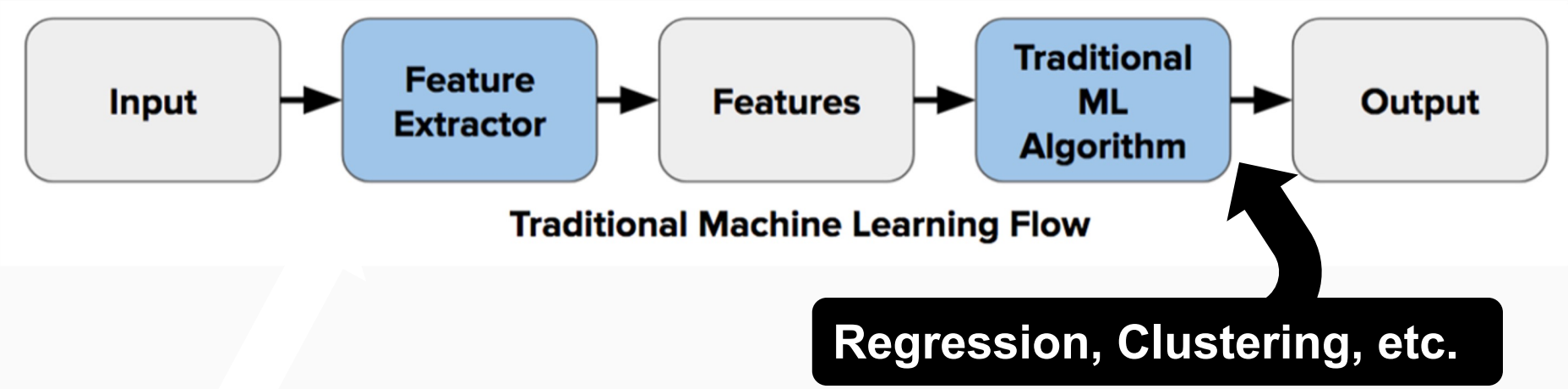




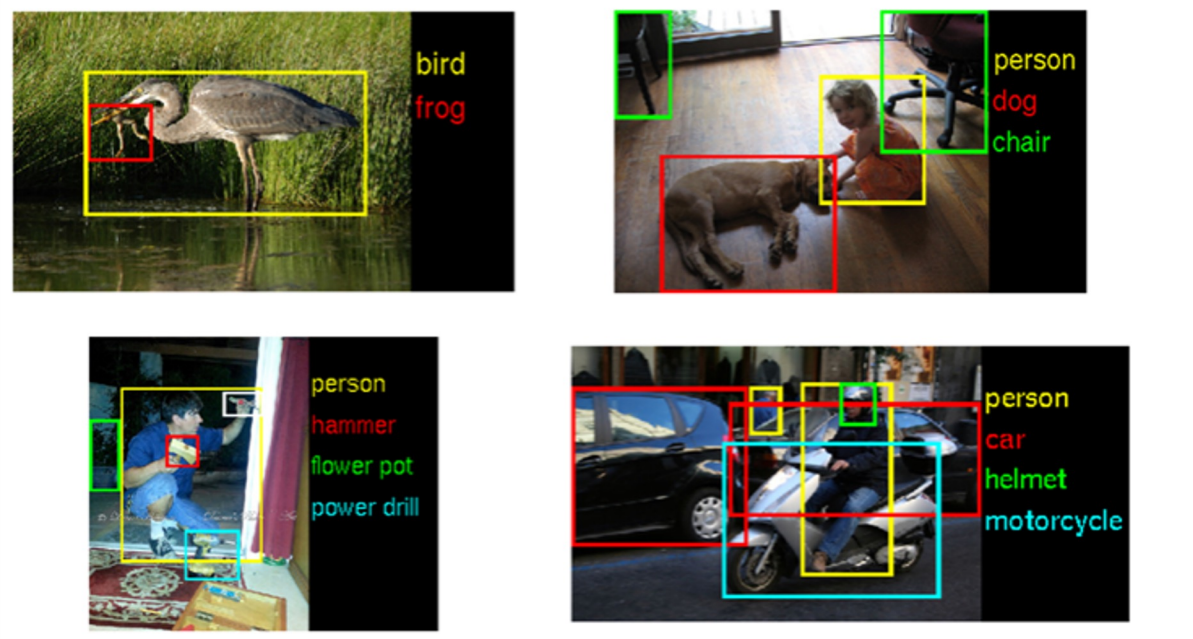


The ImageNet Challenge provided 1.2 million examples of 1,000 **labeled** items and challenged algorithms to learn from the data and then was tested on another 100,000 images

What features are needed for Object Detection?



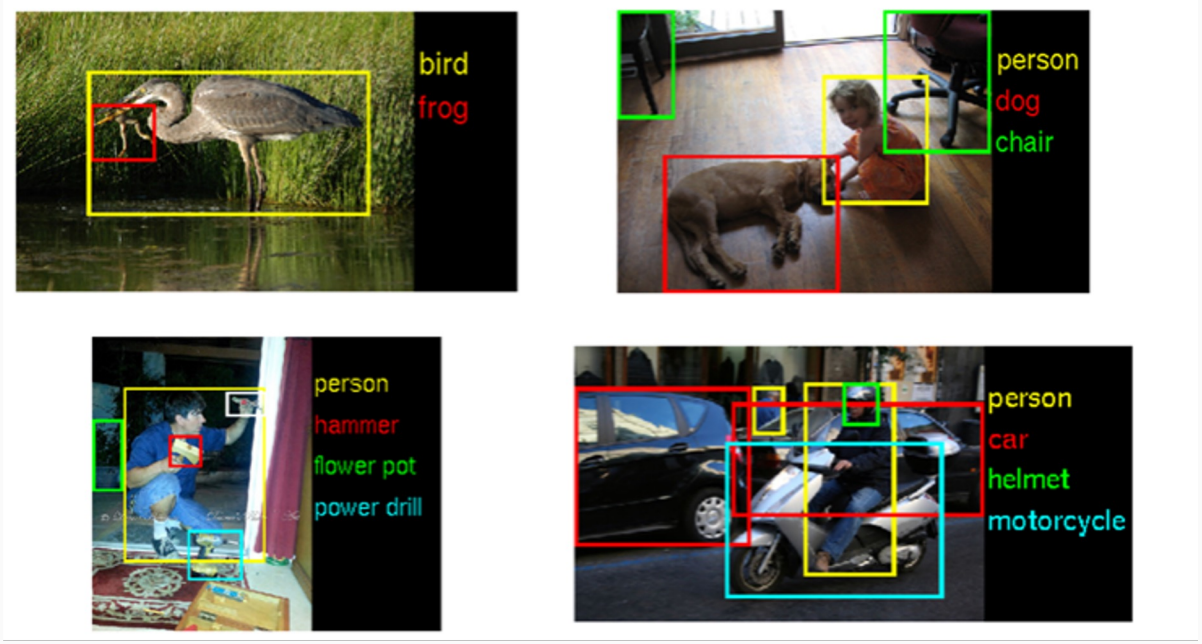
What features are needed for Object Detection?



In 2010 teams had
75-50% error

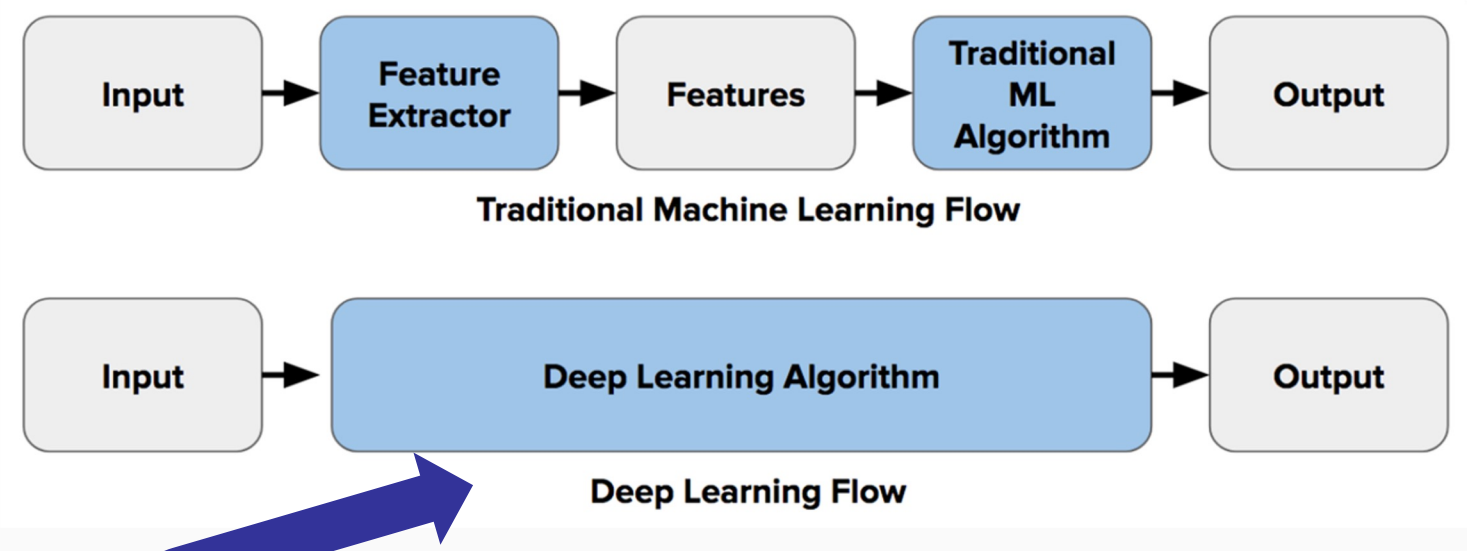
In 2011 teams had
75-25% error

What features are needed for Object Detection?



In 2012 still no team had less than 25% error barrier except **AlexNet at 15%**

What features are needed for Object Detection?

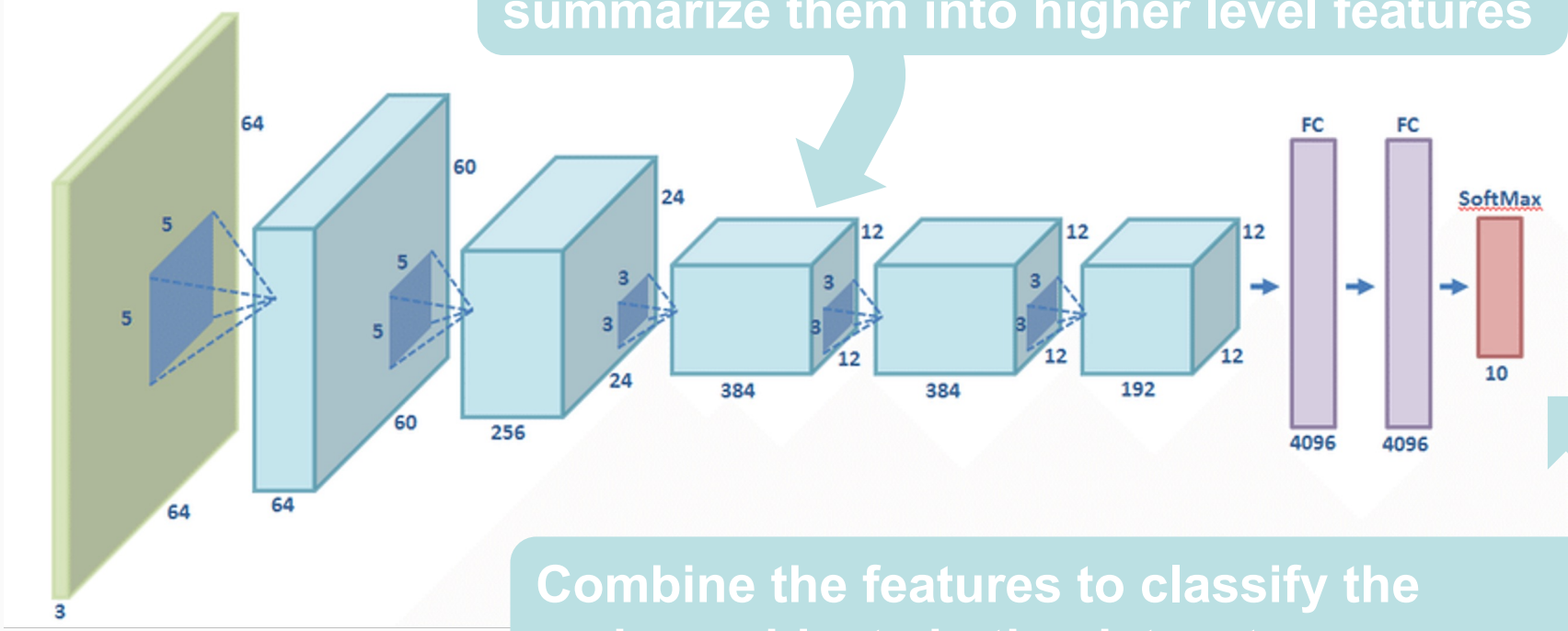


Let the computer figure out its own features and how to combine them!

AlexNet

[AlexNet Paper]

Use convolutions to find features and the summarize them into higher level features



Combine the features to classify the various objects in the dataset

How might we find these features? Convolutions



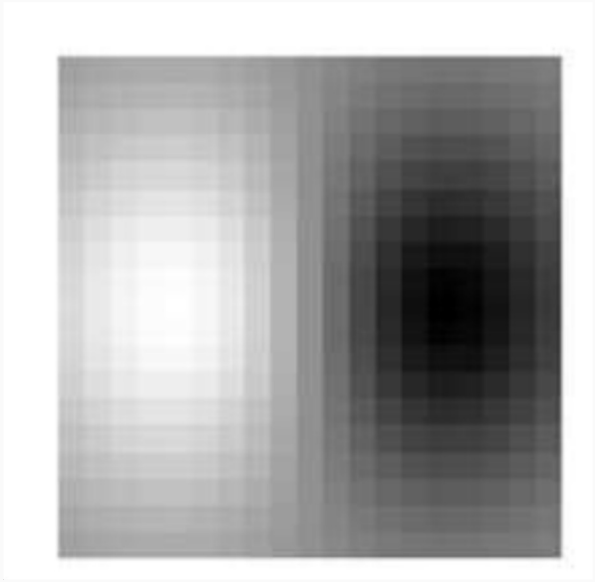
-1	0	1
-2	0	2
-1	0	1

Features

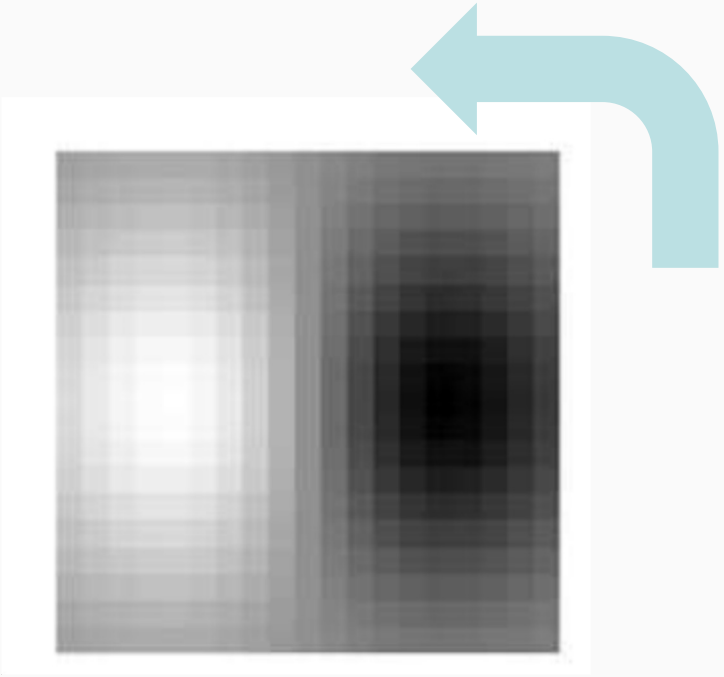


How might we find these features? Convolutions

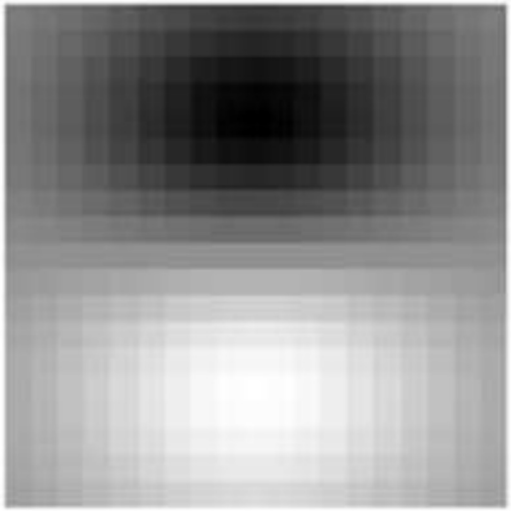
-1	0	1
-2	0	2
-1	0	1



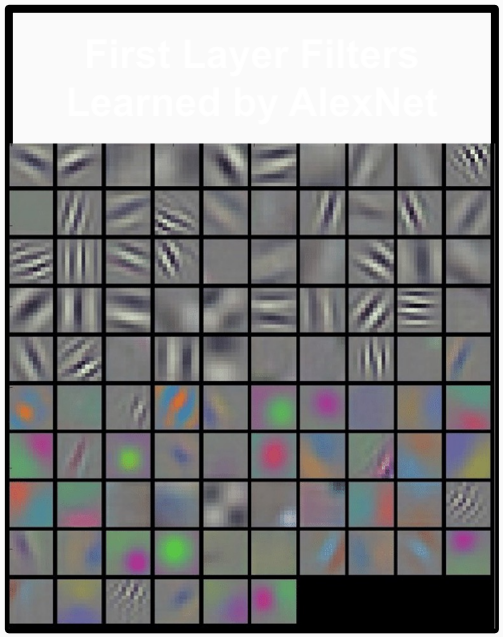
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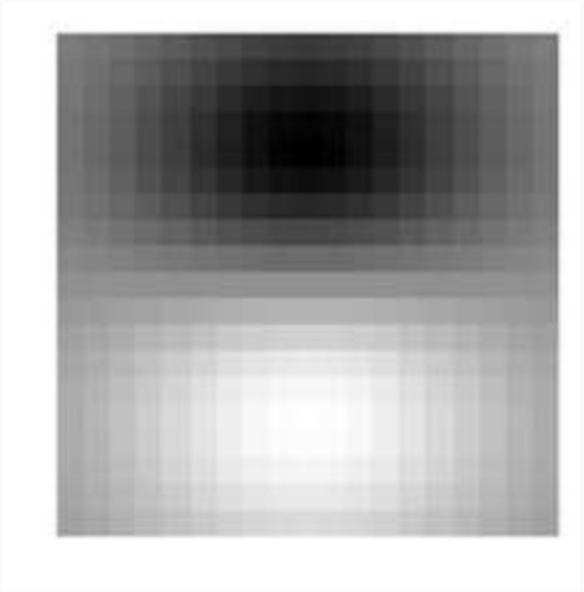


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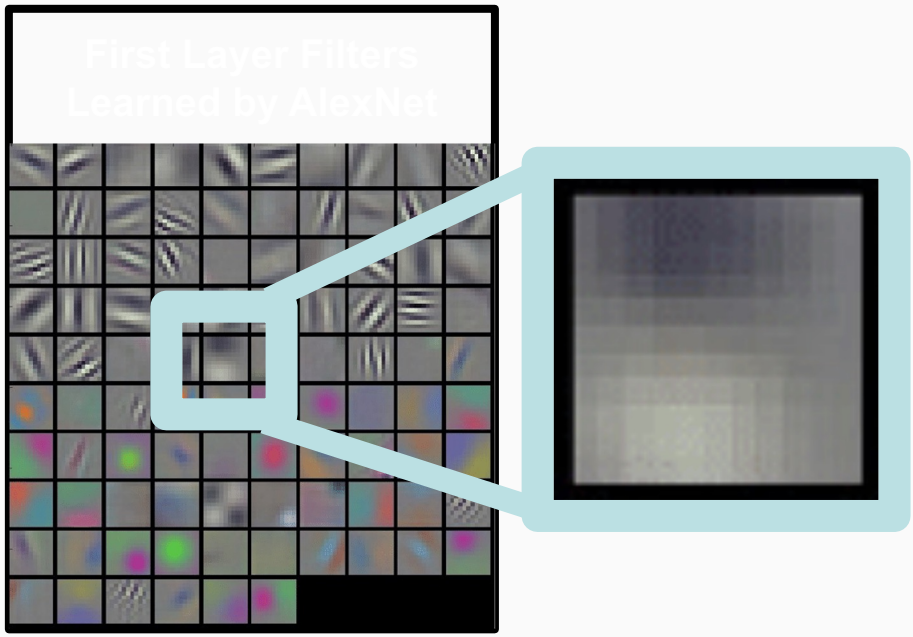


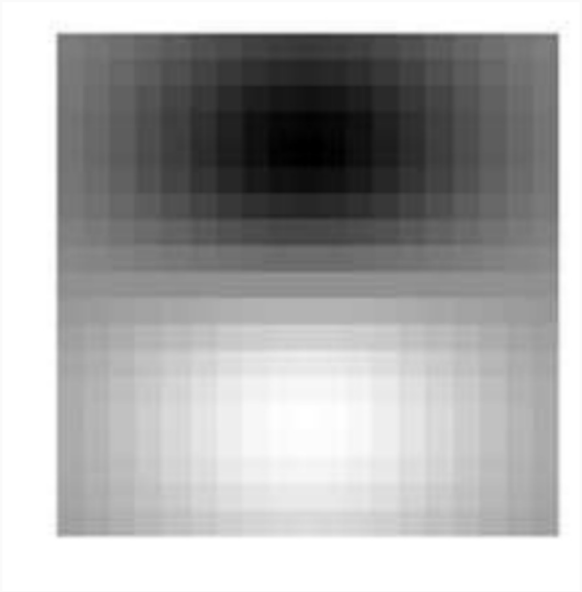
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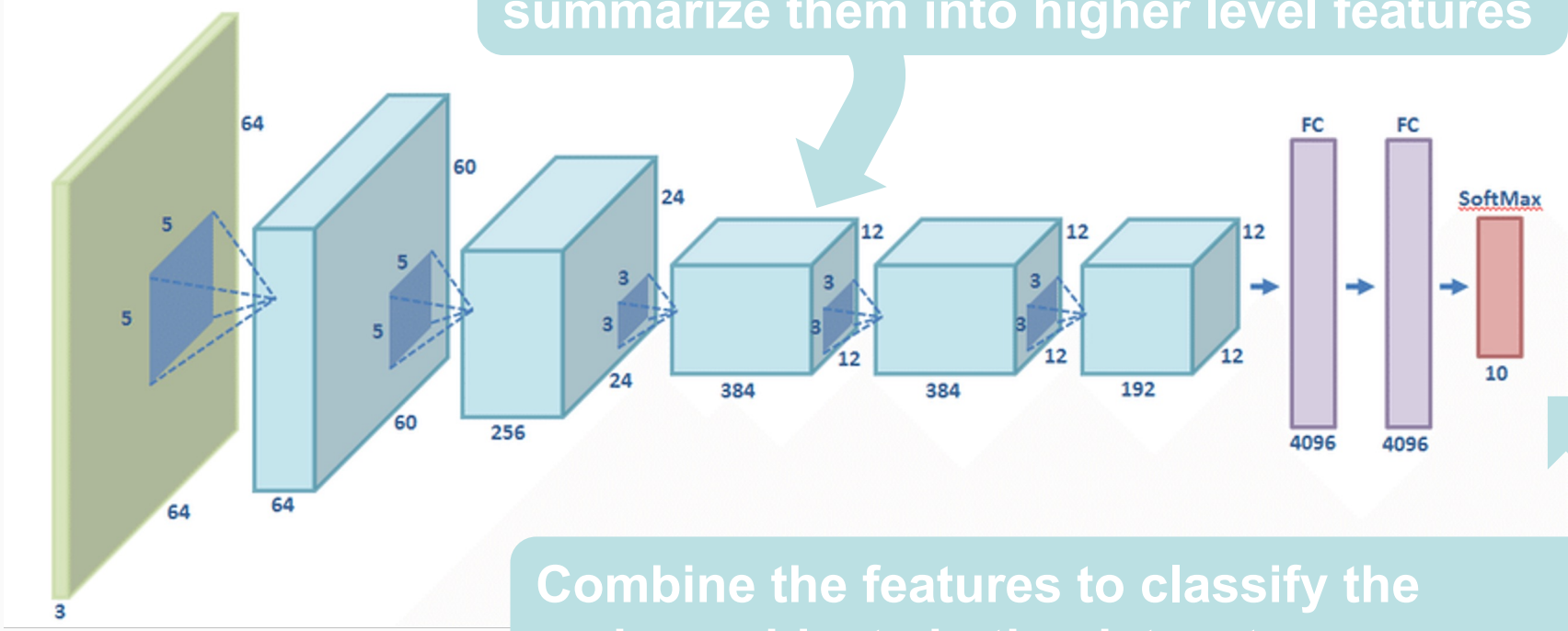




AlexNet

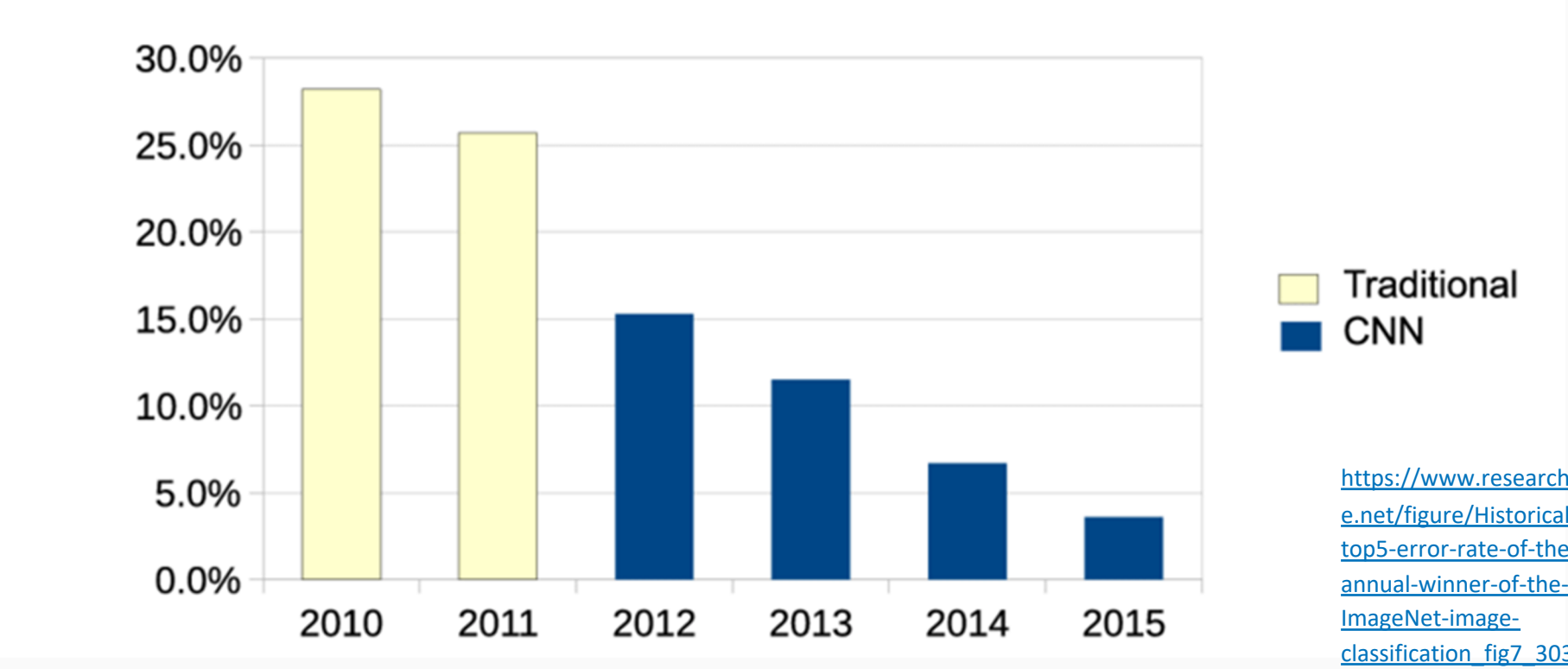
AlexNet Paper

Use convolutions to find features and the summarize them into higher level features



Combine the features to classify the various objects in the dataset

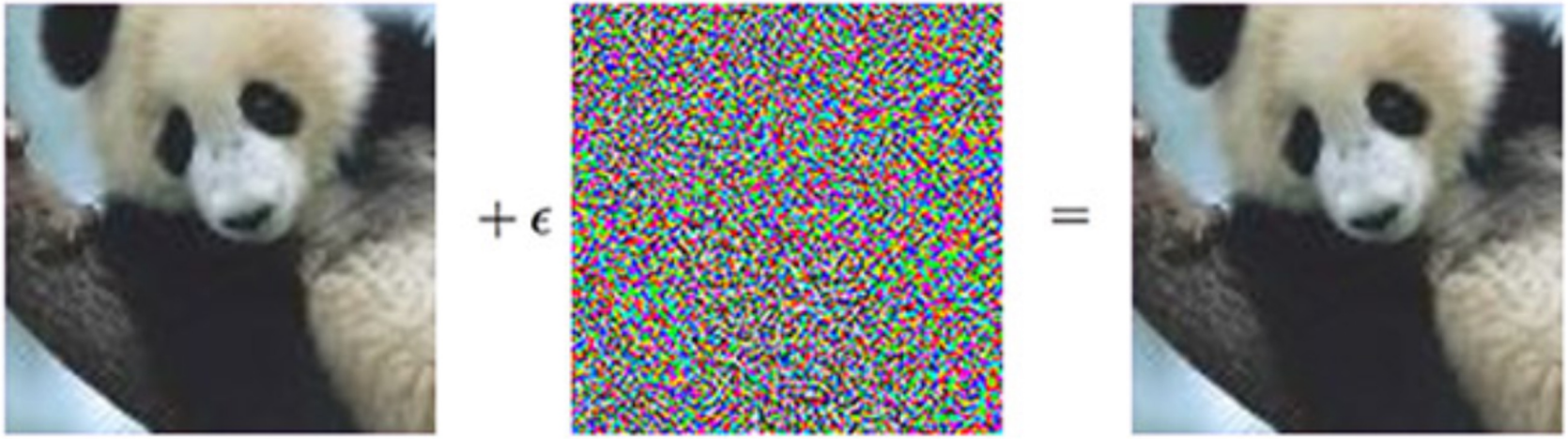
What features are needed for Object Detection?



https://www.researchgate.net/figure/Historical-top5-error-rate-of-the-annual-winner-of-the-ImageNet-image-classification_fig7_303992986

A word of caution...

Ackerman "Hacking the Brain With Adversarial Images"



"panda"
57.7% confidence

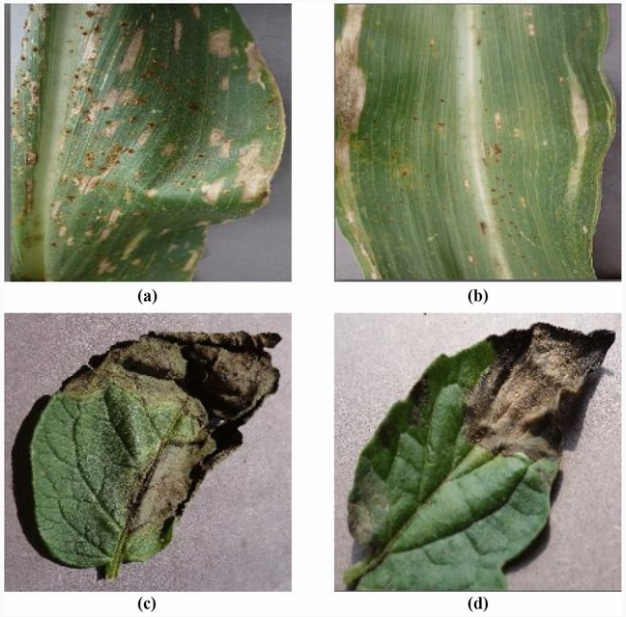
There is **no model** of the world semantically just mathematically

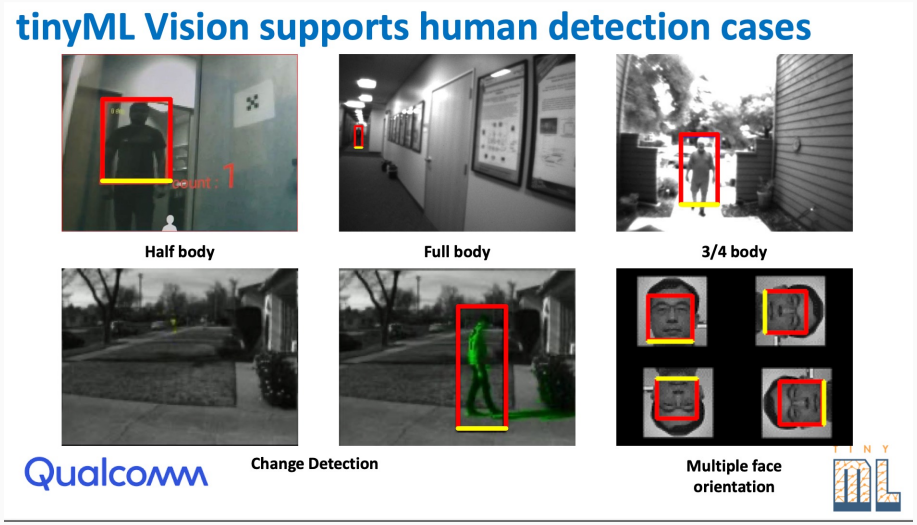
"gibbon"
99.3% confidence

Lab time!

Lab

- Let's get started with the first TinyML Lab!
- **We will work on Image Classification**



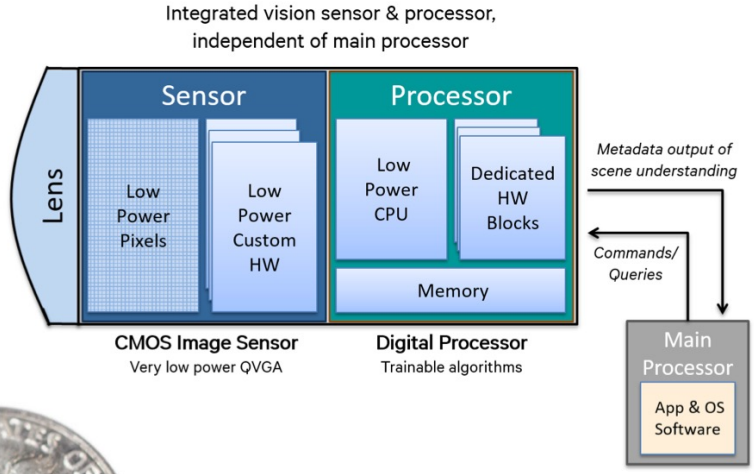


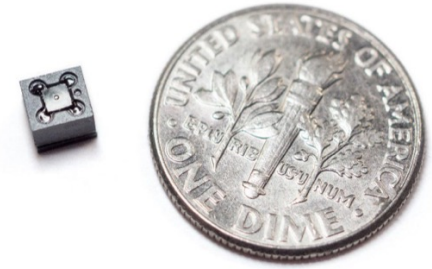
Credit: Evgeni Gousev, tinyML Foundation

tinyML for Always-On Vision
Qualcomm always-on computer vision module

Key features:

- Ultra-low power, < 1 mW (end-to-end)
- Small size
- Privacy (output is metadata)
- Configurable for different use cases
- QVGA sensor, Near-IR compatible
- Low cost





Qualcomm

TINY ML

Credit: Evgeni Gousev, tinyML Foundation

Lab



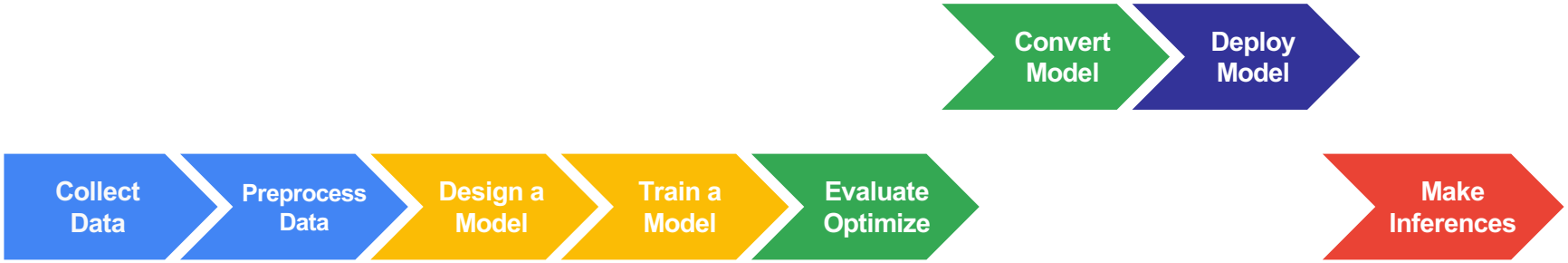
It is better to gather data with the same system that will then run the inference. With image we can also use a smartphone or a PC, but it is better to use the Nicla Vision itself!

Embedded Machine Learning (TinyML) Workflow Review

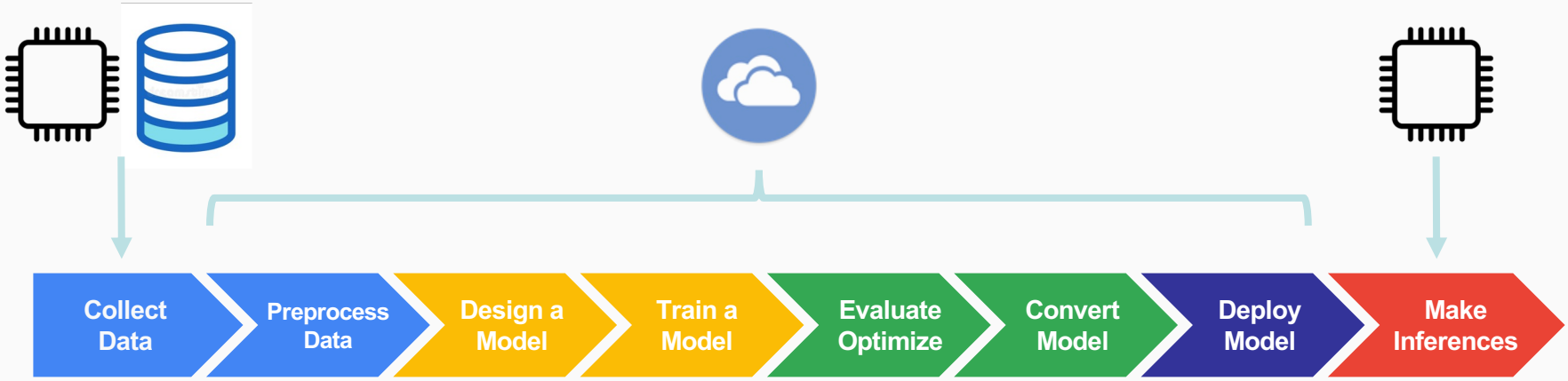
Machine Learning Workflow



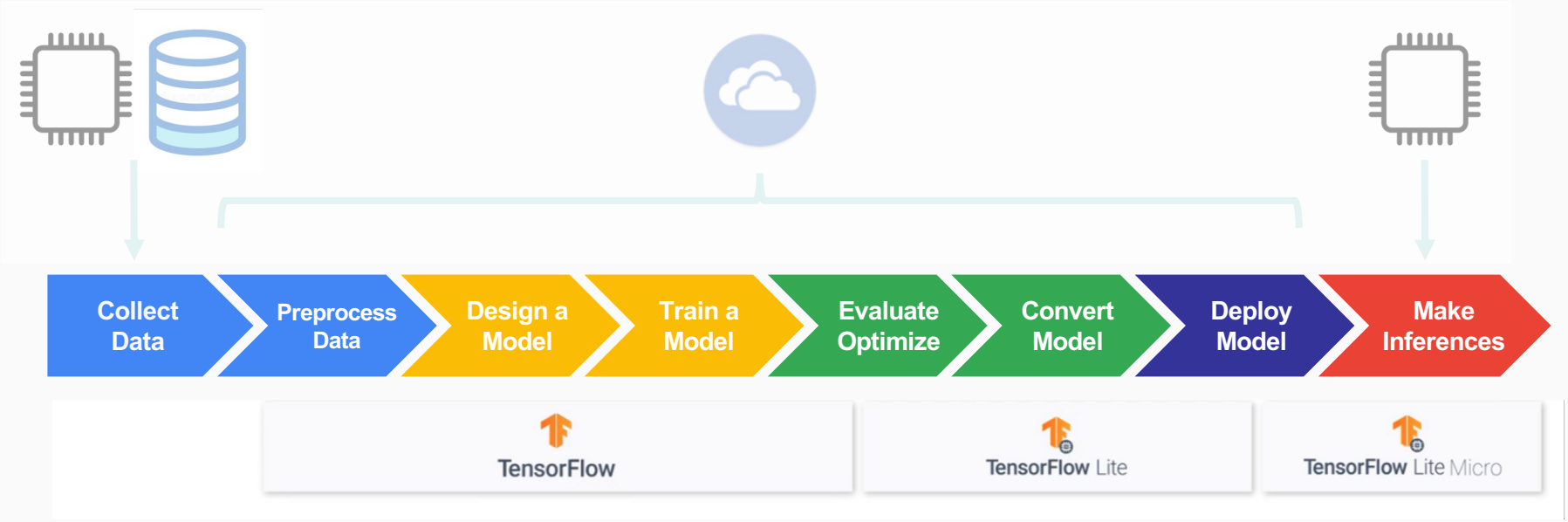
Tiny Machine Learning Workflow (“What”)



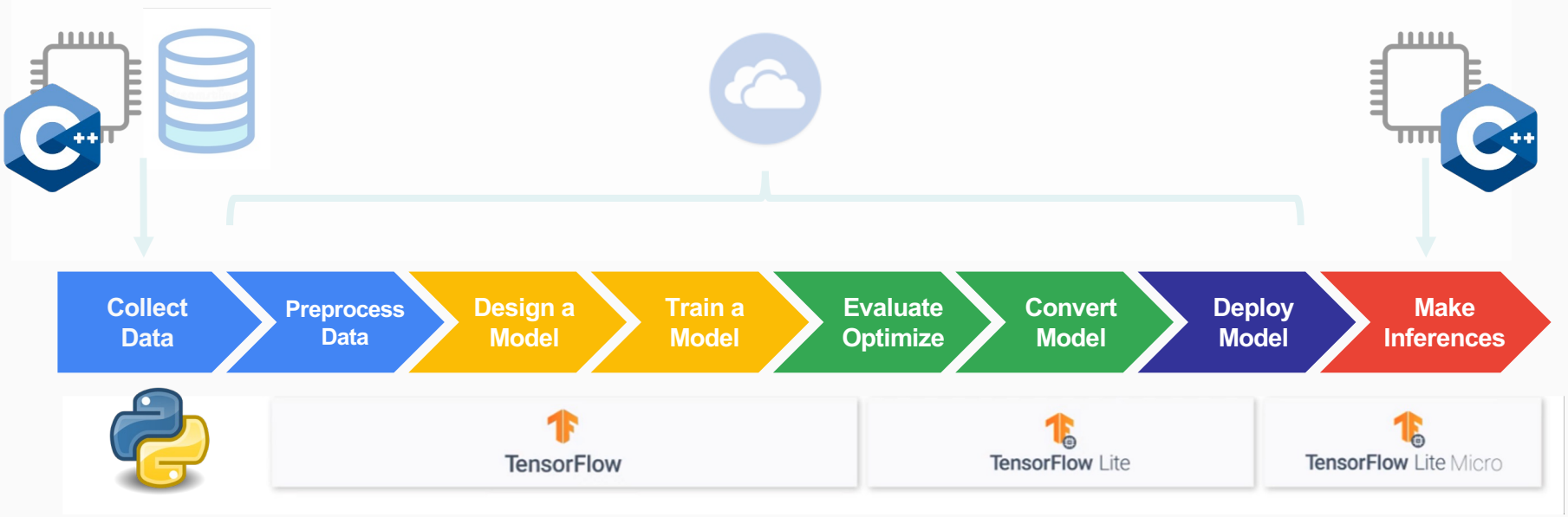
Tiny Machine Learning Workflow ("Where")



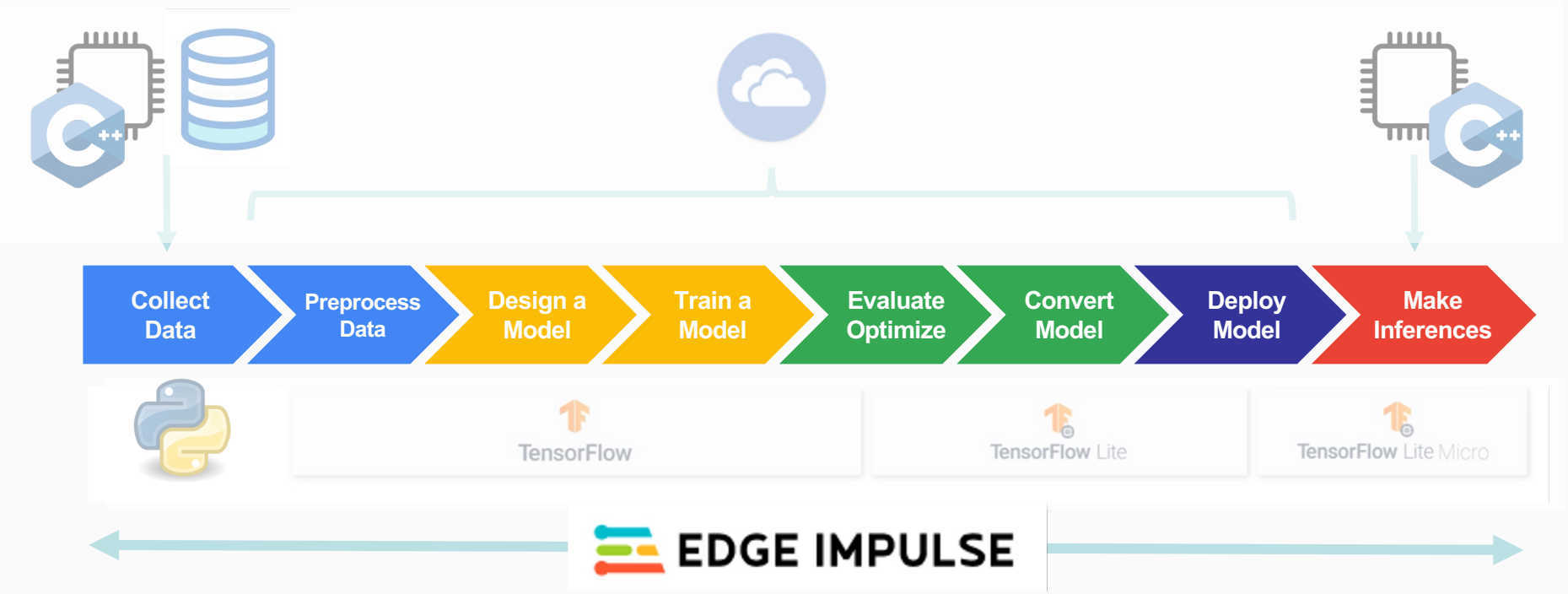
Machine Learning Workflow ("How")

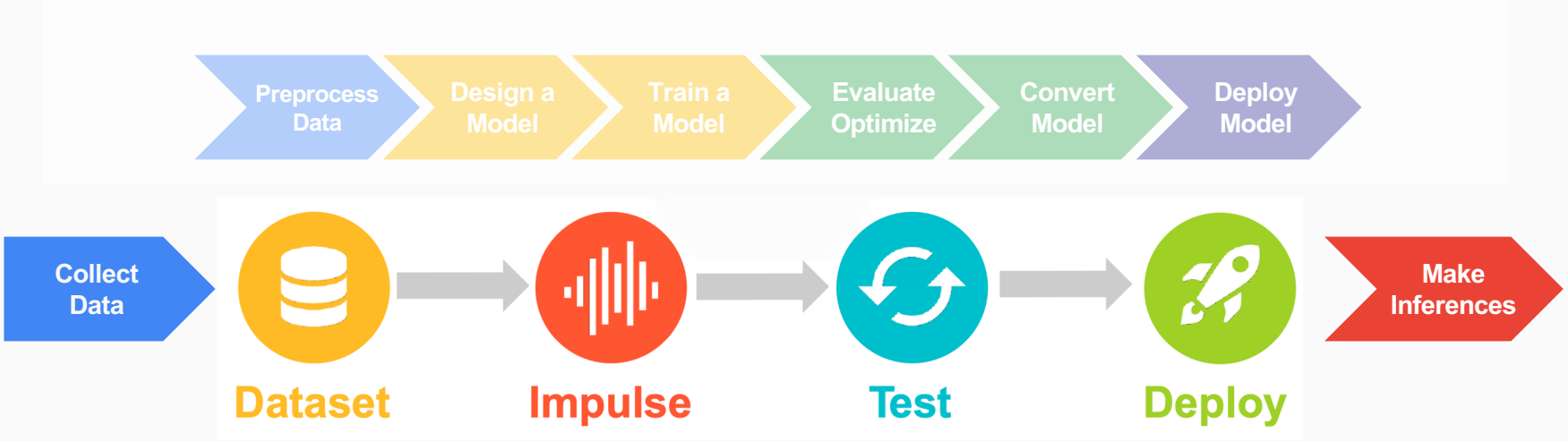


Machine Learning Workflow ("How")

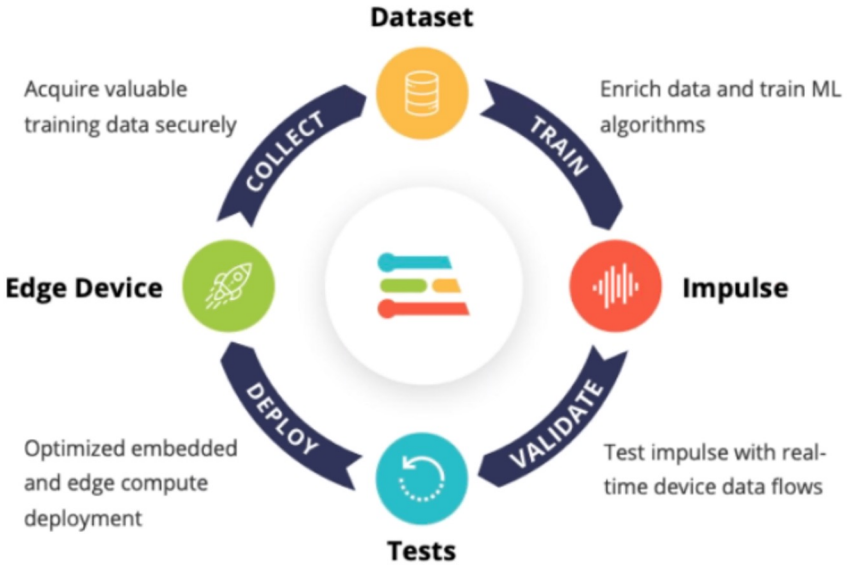


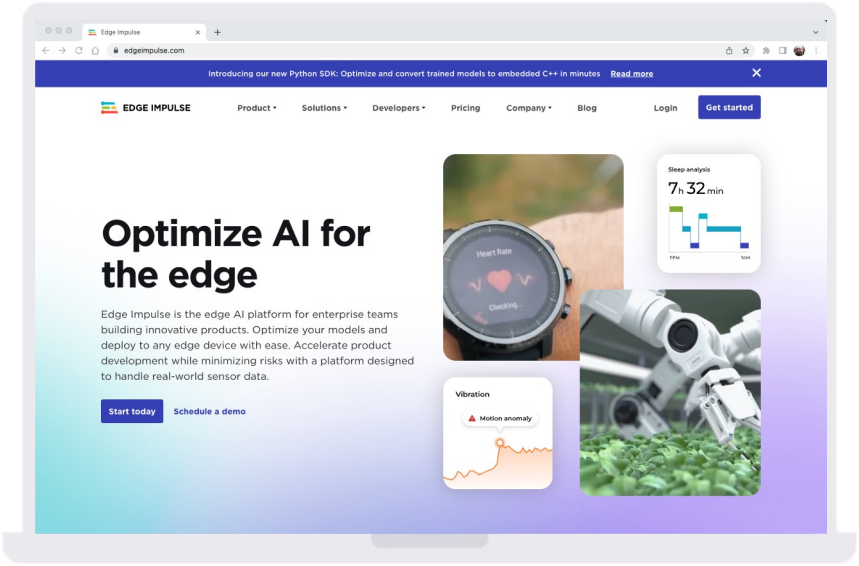
Machine Learning Workflow (“How”)




Preprocess Data
Design a Model
Train a Model
Evaluate Optimize
Convert Model
Deploy Model
Collect Data
Dataset
Impulse
Test
Deploy
Make Inferences

EI Studio - Embedded ML platform ("AutoML")





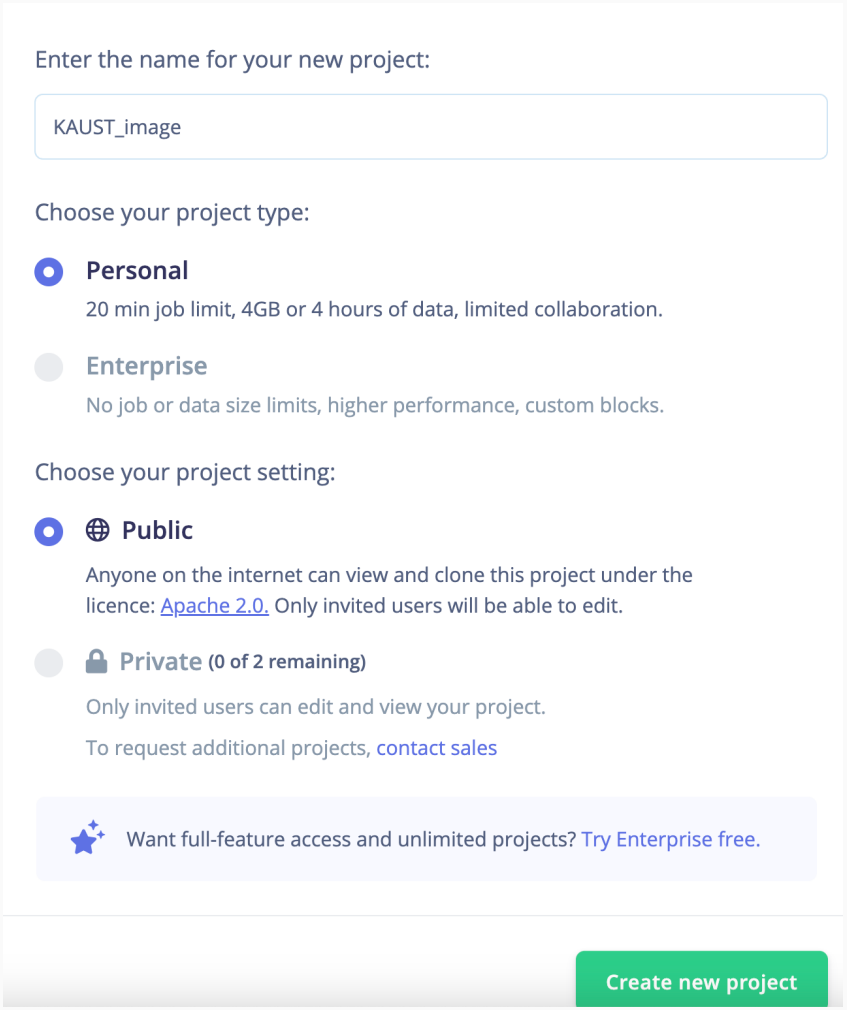
Learn more at http://edgeimpulse.com

[2023 SciTinyML Workshop - Edge Impulse Overview by Shawn Hymel]

Copyright © EdgeImpulse Inc.

Lab

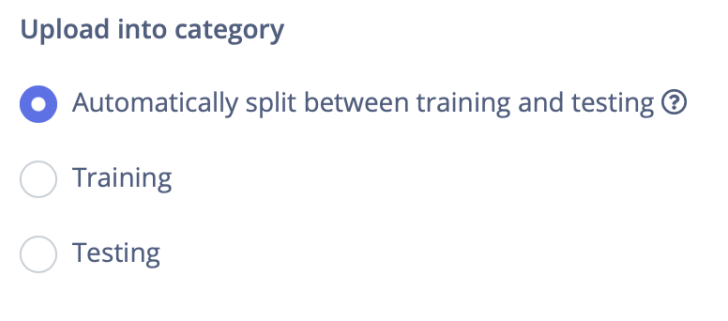
- Register a new account in Edge Impulse. Create **a new project** and give it a name:



Lab

- Once the project is created, select **Add Existing data** and then **Upload data**. Choose the files to be uploaded.

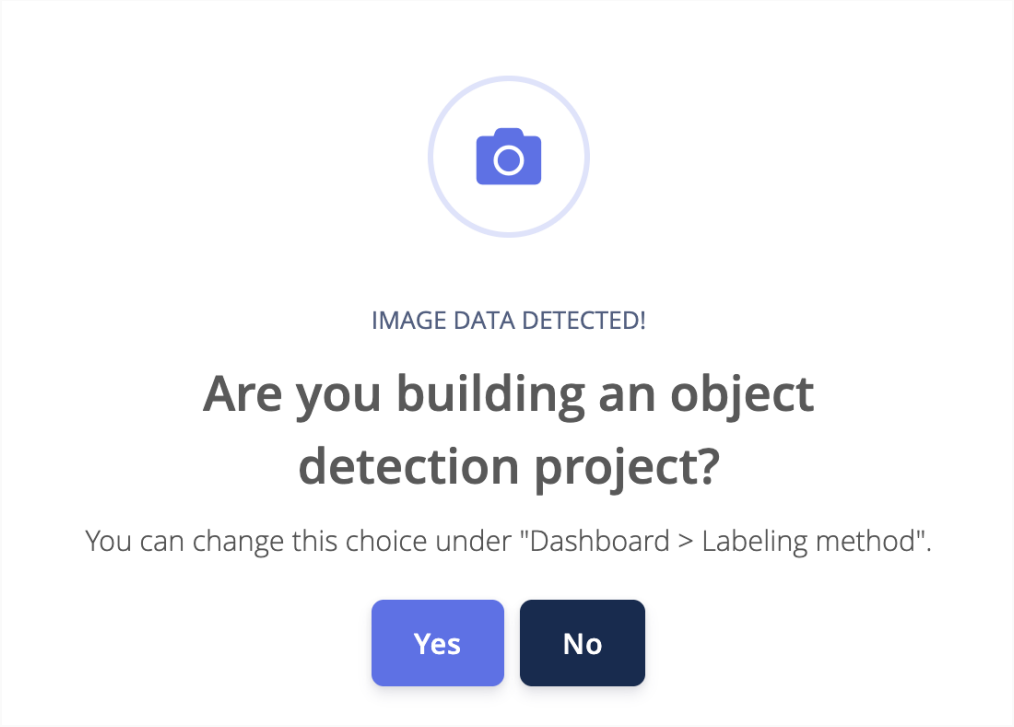
- Leave the default setting and make sure that Split between training and testing is selected.



- Click on Upload data on the lower right of the window.

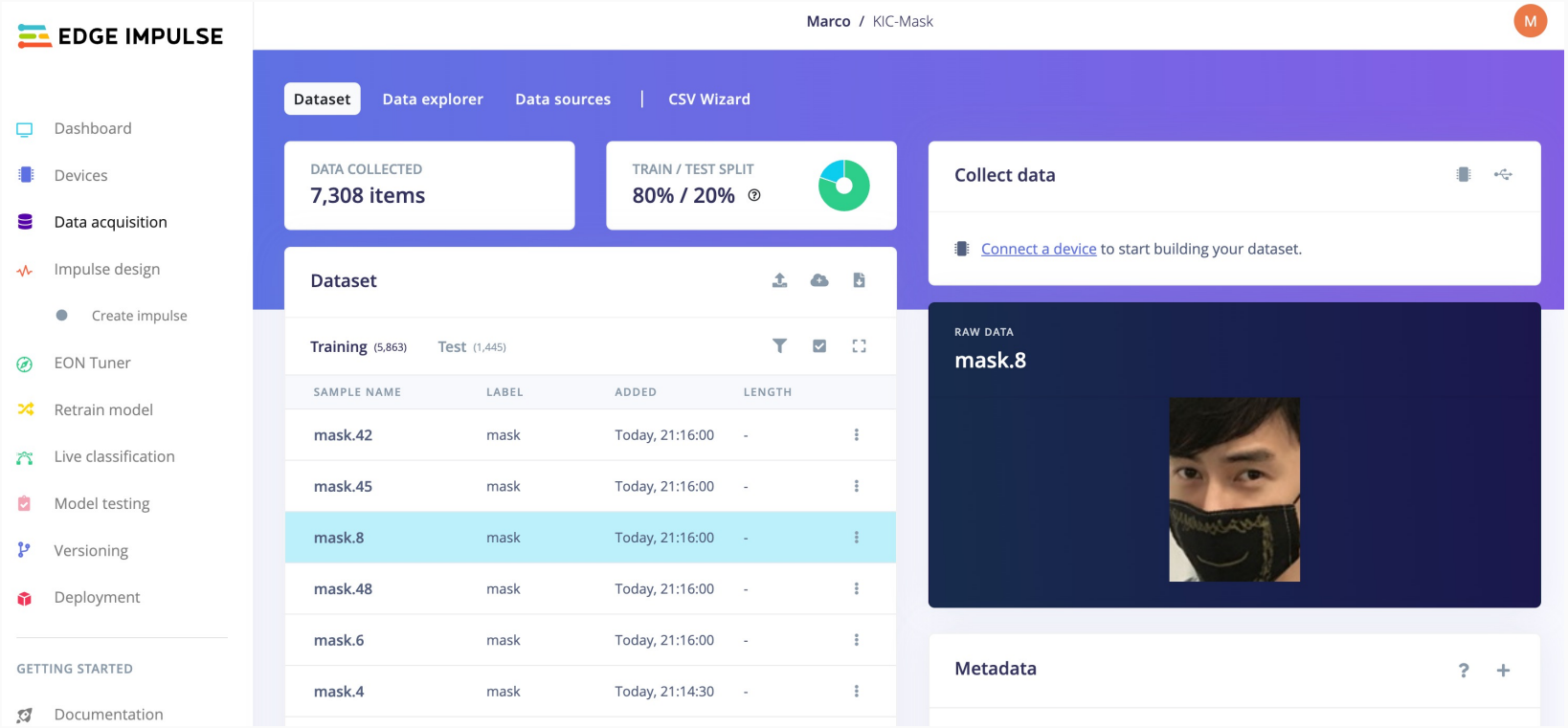
Lab

- If you are working on an image classification project, make sure you click on No in the pop-up page:



Lab

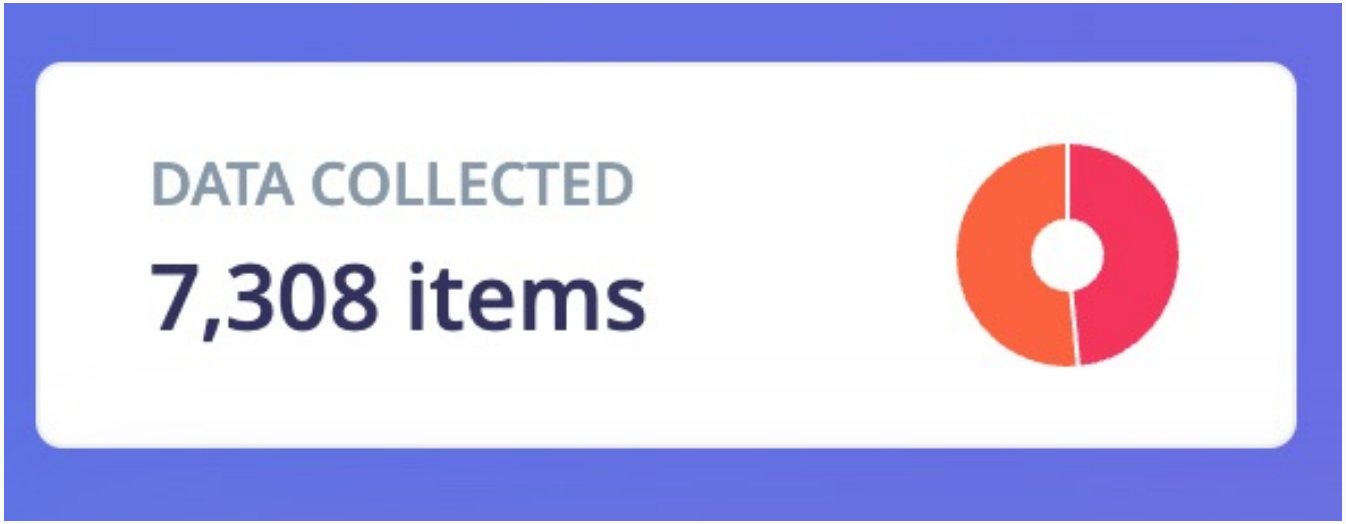
- The uploaded samples will now appear in the Data acquisition section.



Lab



It is important for the dataset to be balanced!



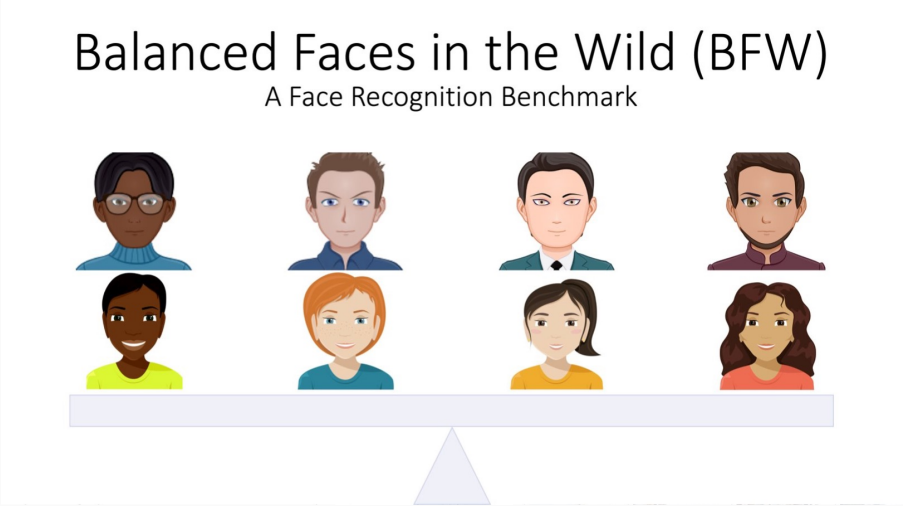
Balanced datasets

- Classification models attempt to categorize data into different buckets. In an imbalanced dataset, one bucket makes up a large portion of the training dataset (the majority class), while the other bucket is underrepresented in the dataset (the minority class). **The problem with a model trained on imbalanced data is that the model learns that it can achieve high accuracy by consistently predicting the majority class, even if recognizing the minority class is equal or more important when applying the model to a real-world scenario.**
- Consider the case of collecting training data for a model predicting medical conditions. Most of the patient data collected, let's say 95 percent, will likely fall into the healthy bucket, while the sick patients make up a much smaller portion of the data. During training, the classification model learns that it can achieve 95 percent accuracy if it predicts "healthy" for every piece of data it encounters. That's a huge problem because what doctors really want the model to do is identify those patients suffering from a medical condition.

https://encord.com/blog/an-introduction-to-balanced-and-imbalanced-datasets-in-machine-learning/#:~:text=Balancing%20a%20dataset%20makes%20training,becoming%20biased%20towards%20o

Balanced datasets

- **Imbalanced datasets** create challenges for predictive modeling, but they're a common and anticipated problem because the real world is full of imbalanced examples.



- **Balancing a dataset** makes training a model easier because it helps prevent the model from becoming biased towards one class. In other words, the model will no longer favor the majority class just because it contains more data.

Balanced datasets

Unbalanced Dataset

Cats



Dogs



Balanced datasets

Get More Data

Cats



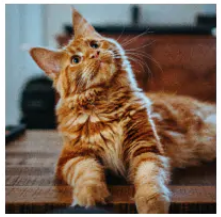
Dogs



Balanced datasets

Undersampled

Cats



Dogs



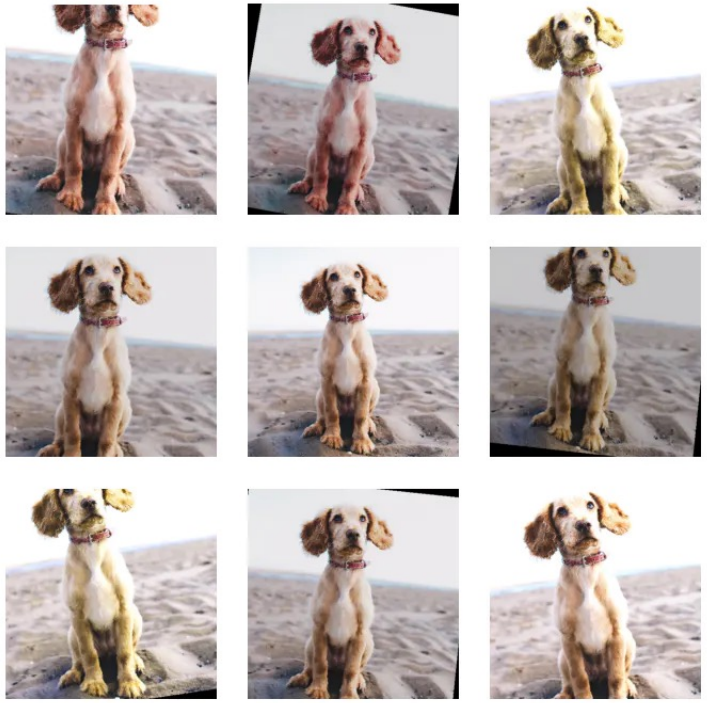
Balanced datasets

Oversampled

Cats



Dogs



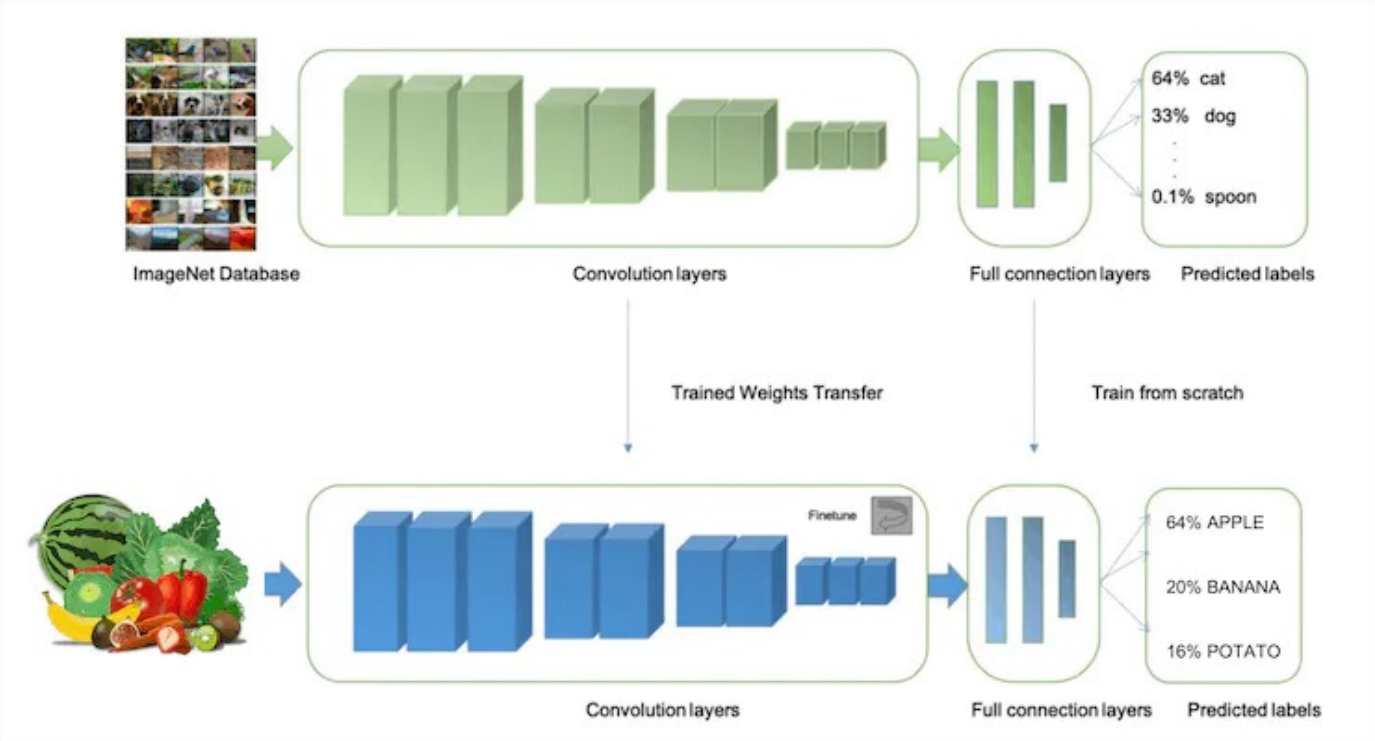
Lab

- An impulse takes raw data (in this case, images), extracts features (resize pictures), and then uses a learning block to classify new data.

- Classifying images is the most common use of Deep Learning, but much data should be used to accomplish this task. Do we have enough images for each category? Not at all! We will need thousand of images to train the model.

- We can solve this issue by re-training a previously trained model with thousands of images. We called this technique **Transfer Learning (TL)**.

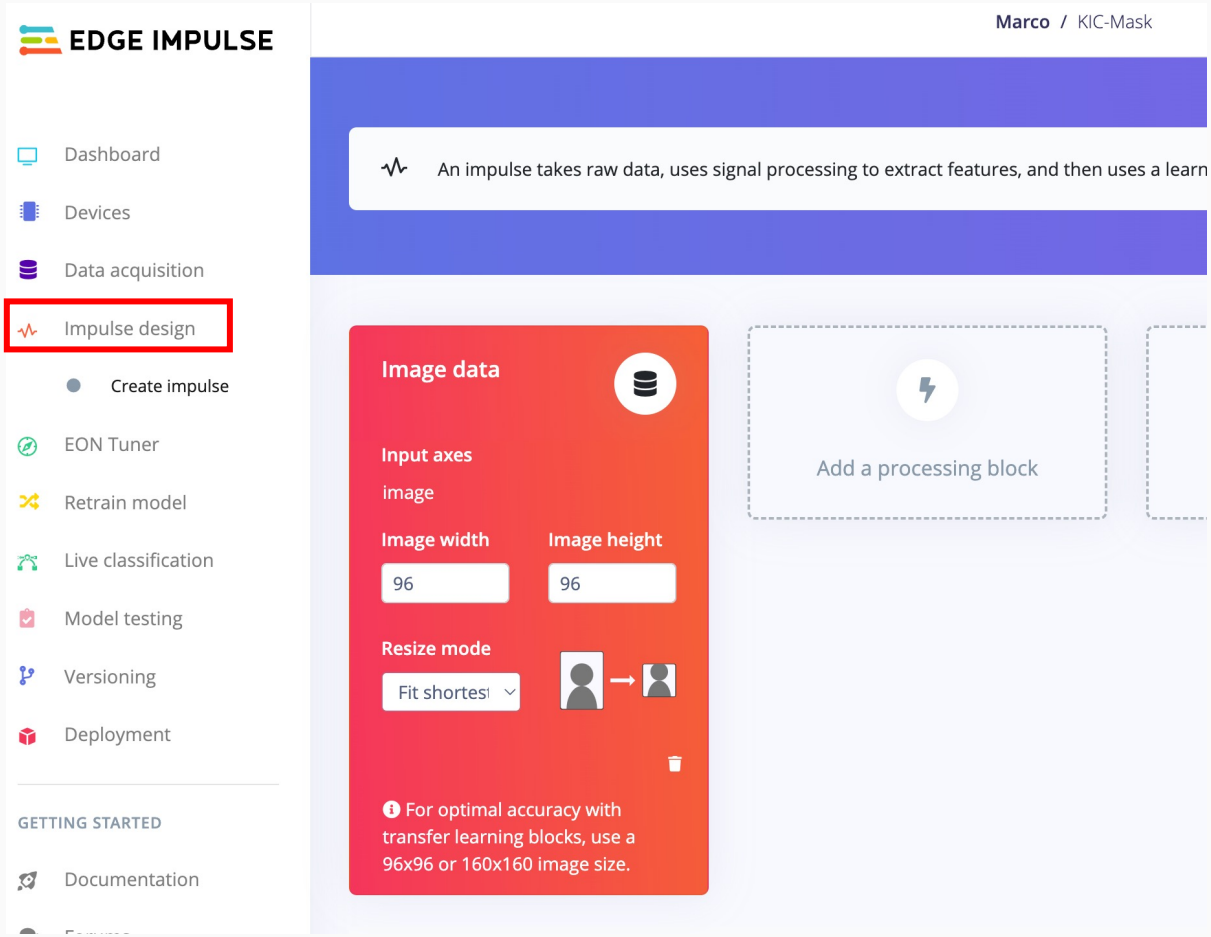
Lab



With TL, we can fine-tune a pre-trained image classification model on our data, performing well even with relatively small image datasets (our case).

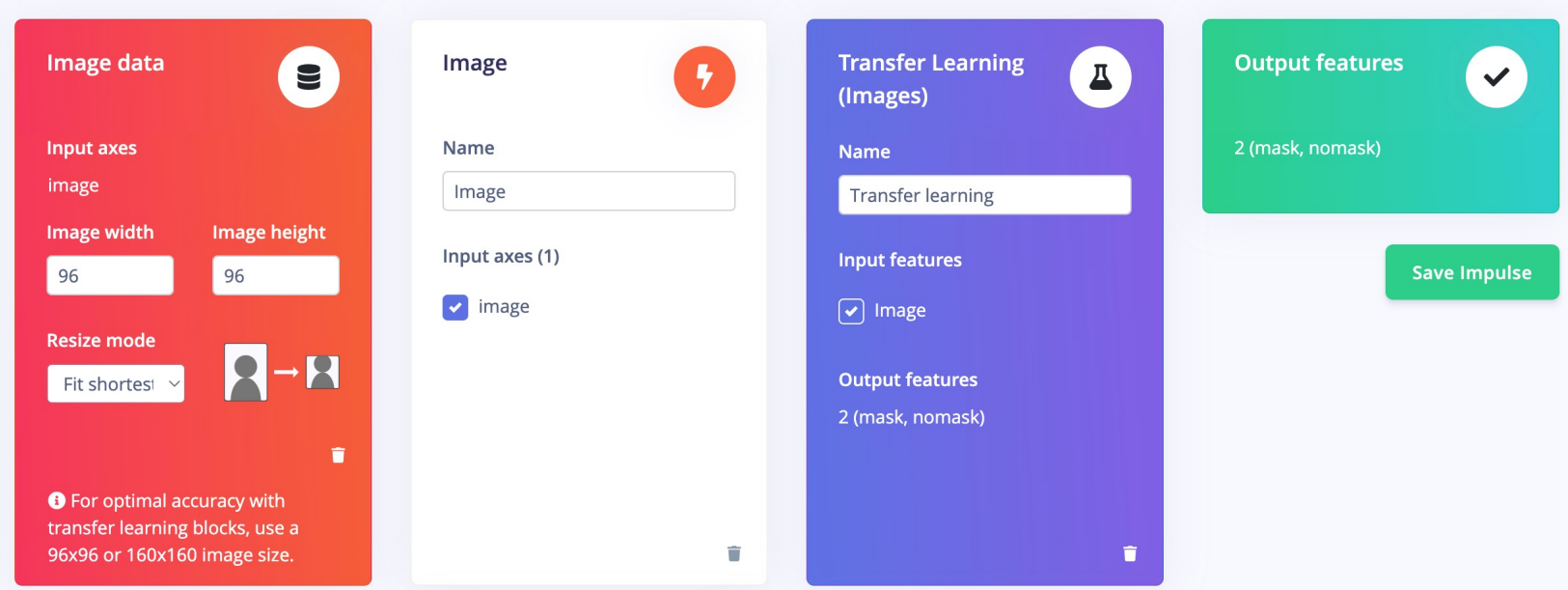
Lab

- Let's start creating the impulse. Click on **Impulse Design**.



Lab

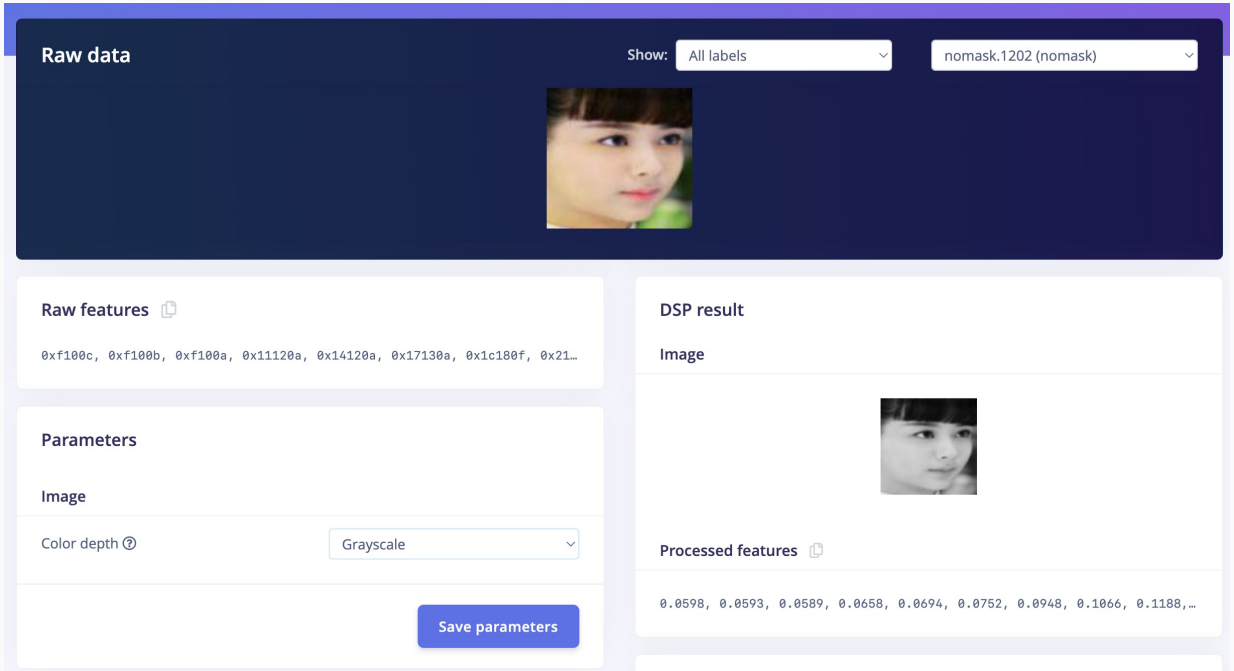
- Starting from the raw images, we will resize them (96x96) pixels and use Transfer Learning.
- Click on Save Impulse.



Lab

- By changing the data samples to Greyscale, we save a lot of memory. Each sample has now 96x96x1 features (216). Using RGB color depth, the dimension would be three times bigger. Working with Grayscale helps to reduce the amount of final memory needed for inference.

- Click on Image, on the left, to choose Greyscale. Click on Save parameters.



Lab

- In 2007, Google introduced **MobileNetV1**, a family of general-purpose computer vision neural networks designed with mobile devices in mind to support classification, detection, and more. MobileNets are small, low-latency, low-power models parameterized to meet the resource constraints of various use cases.

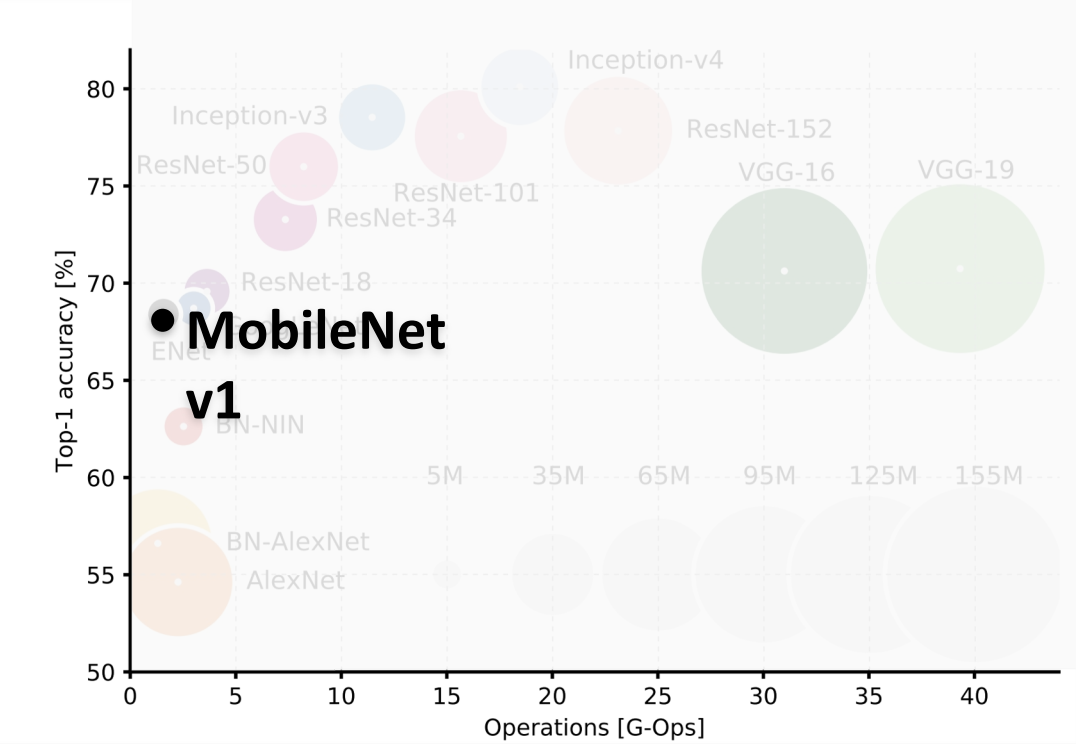
- Although the base MobileNet architecture is already tiny and has low latency, many times, a specific use case or application may require the model to be smaller and faster. MobileNet introduces a straightforward parameter **α** (alpha) called **width multiplier** to construct these smaller and less computationally expensive models. The role of the width multiplier α is to thin a network uniformly at each layer.

Lab

- Edge Impulse Studio offers MobileNet V1 (96x96 images) and V2 (96x96 and 160x160 images), with several different α values (from 0.05 to 1.0). For example, you will get the highest accuracy with V2, 160x160 images, and α=1.0. Of course, there is a trade-off. The highest the accuracy, the more memory (around 1.3M RAM and 2.6M ROM) will be needed to run the model and imply more latency.

- The smaller footprint will be obtained at another extreme with **MobileNet V1 and α=0.10 (around 53.2K RAM and 101K ROM).**

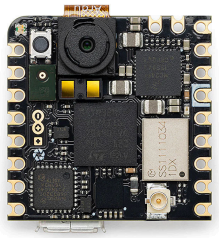
Model Evolution



MobileNet v1

Model	Size	Top-1 Accuracy
MobileNet v1	***16 MB***	0.713

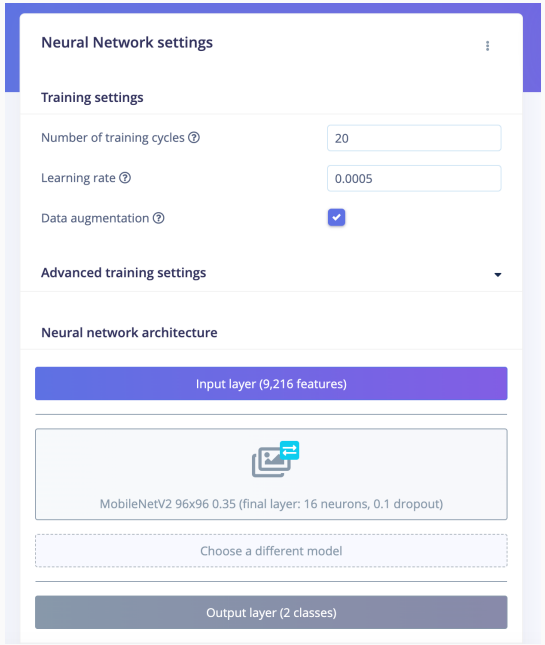
Fine for mobile phones with GB of RAM, but 64X microcontroller RAM



Our board only has **1MB** of RAM (memory)

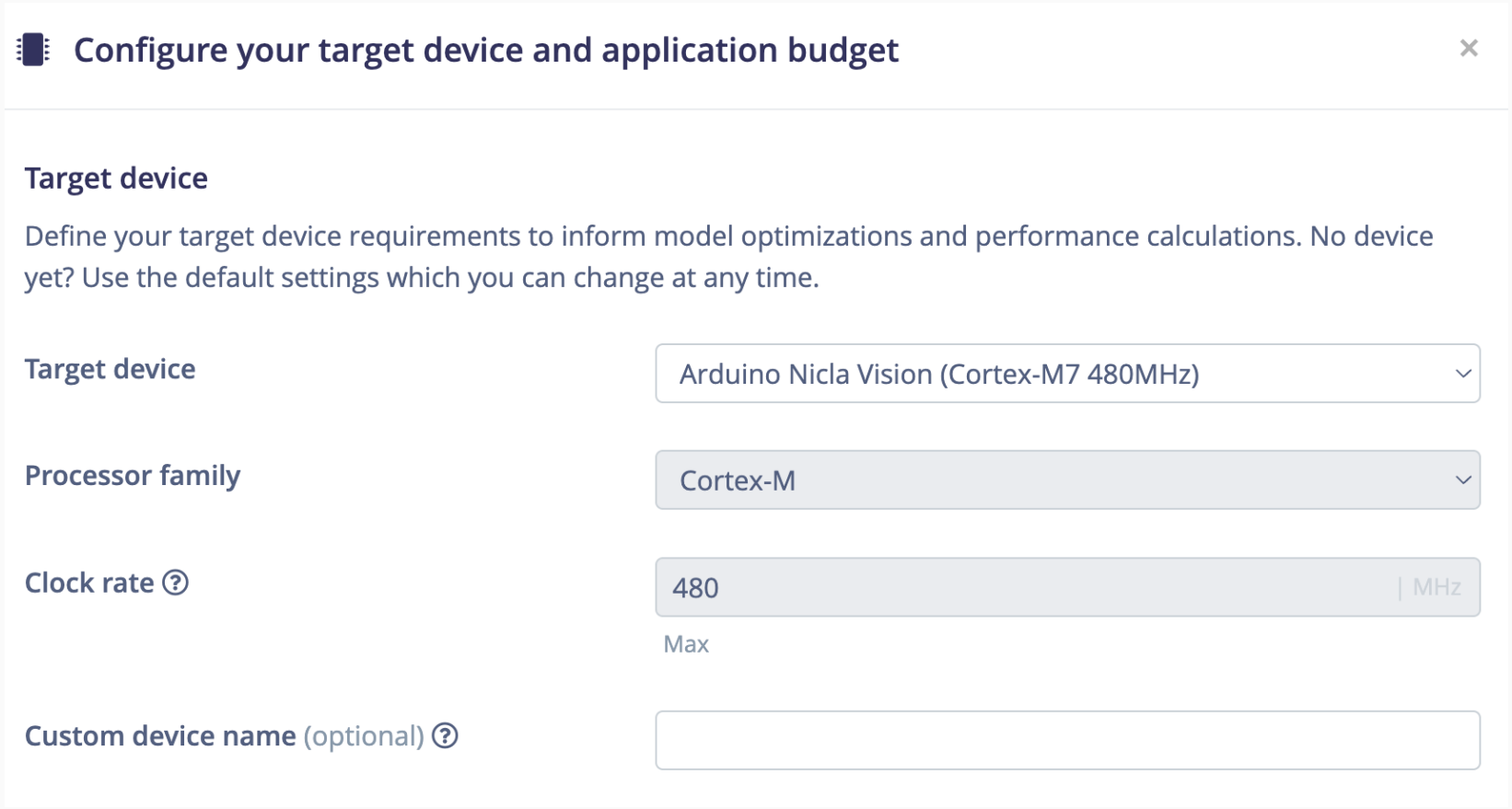
Lab

- Another important technique to be used with Deep Learning is **Data Augmentation**. Data augmentation is a method that can help improve the accuracy of machine learning models, creating additional artificial data. A data augmentation system makes small, random changes to your training data during the training process (such as flipping, cropping, or rotating the images).
- Let's choose Transfer Learning on the left, and start with the default values (MobileNetV2, 96x96, 0.35). Don't forget to select **Data augmentation.**



Lab

- Before clicking on Start Training, choose on the top right side of the window the **Target device**. Click on Nicla Vision and then on **Start Training**.



Lab



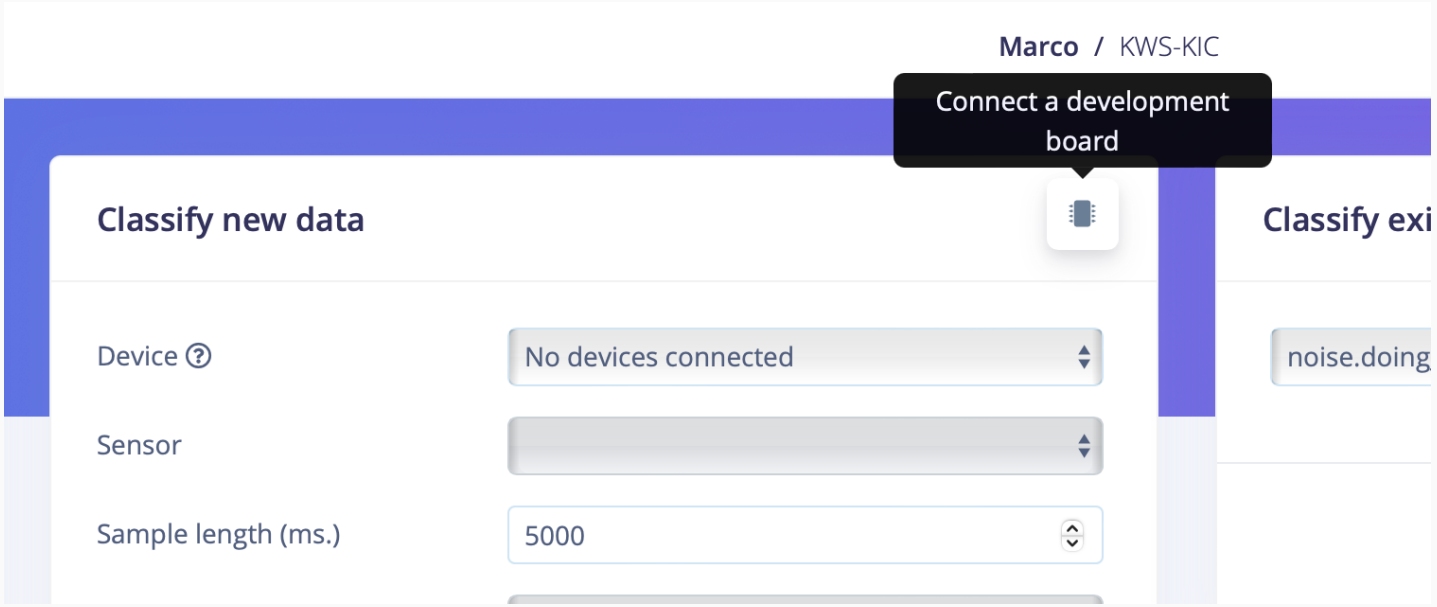
Lab

- The result is great. Testing the model with the data put apart before training (Test Data), we got an accuracy of approximately 97%.



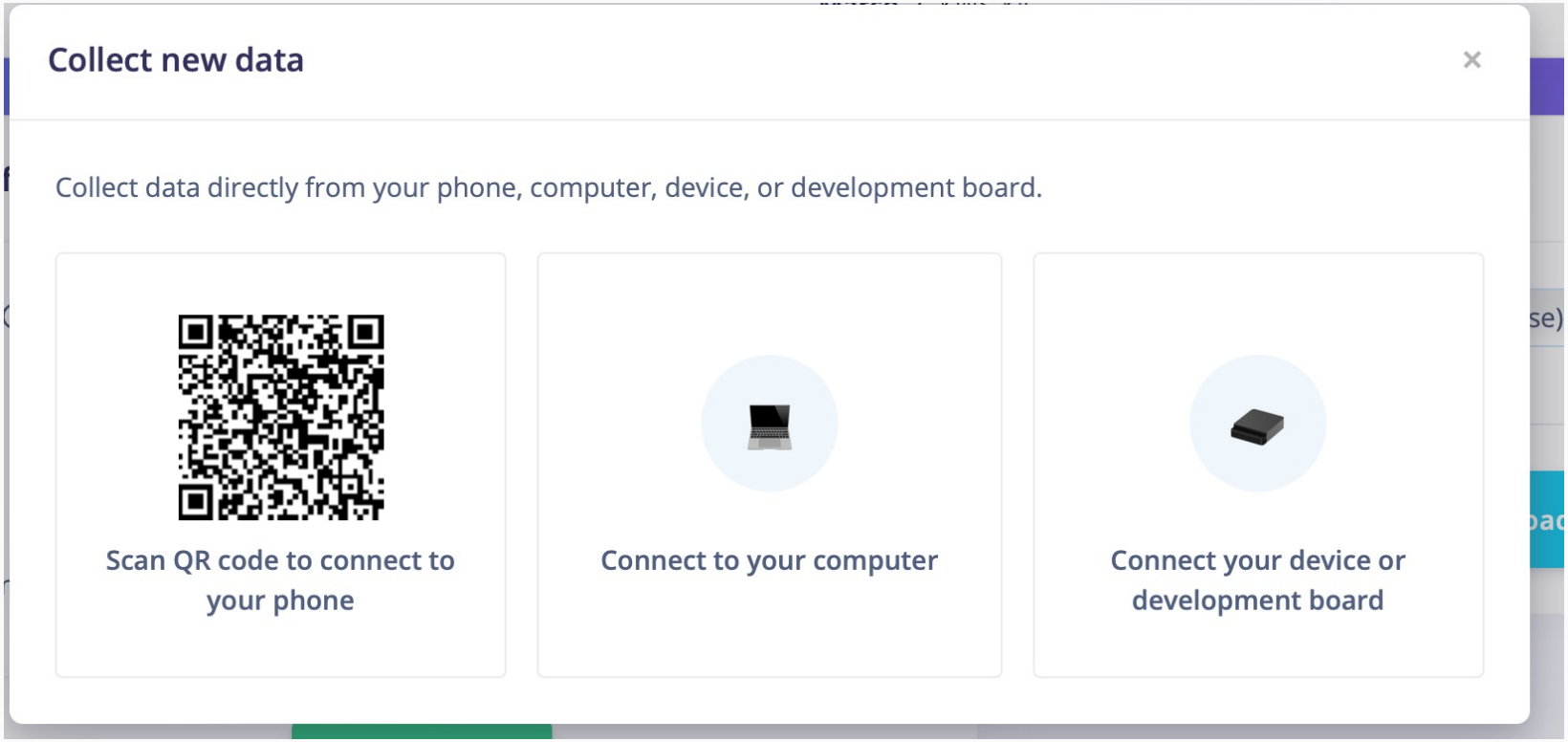
Lab

- Before deployment of the model on the Nicla device, it is possible to perform Live Classification using a smartphone. Go to the **Live Classification** section and click on **Connect a Development board**:



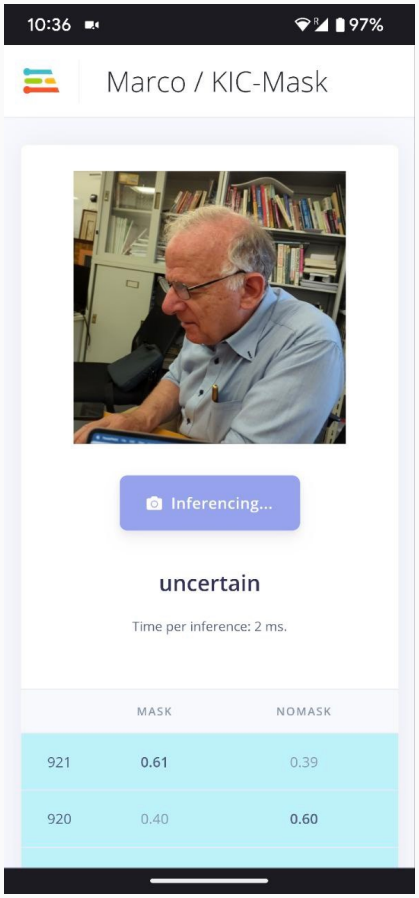
Lab

- Point your phone to the QR code and select the link.



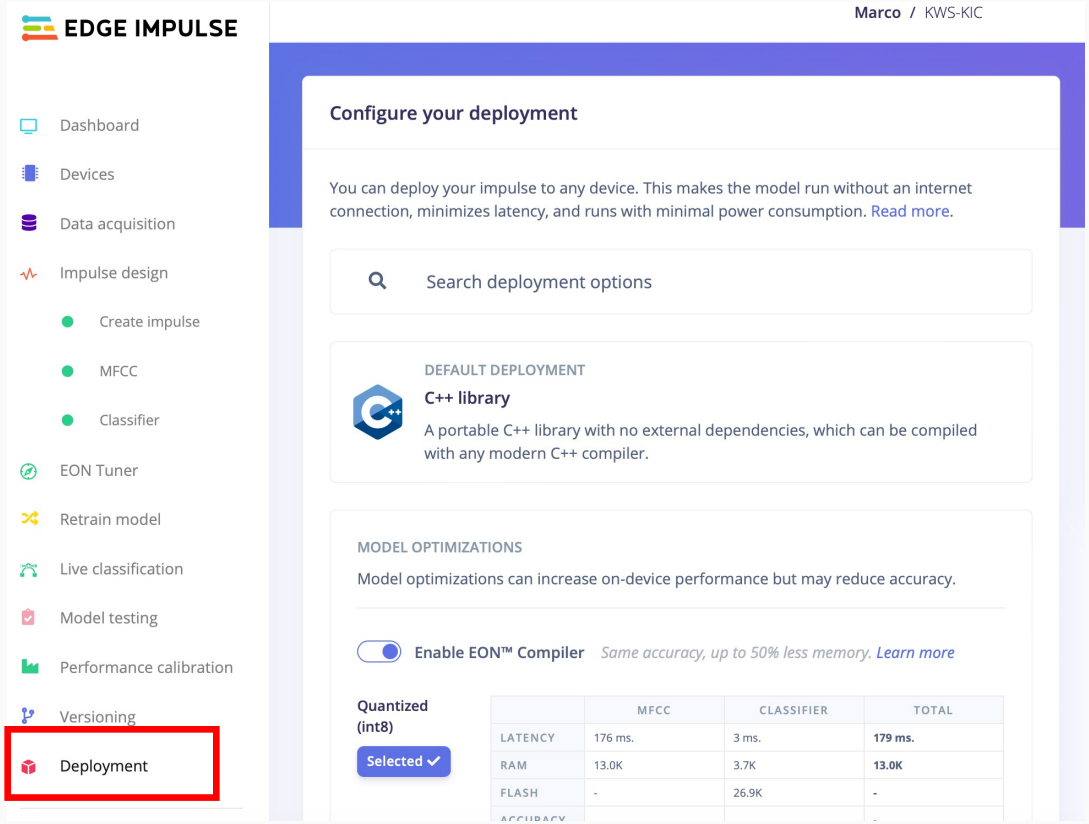
Lab

- Your phone will be connected to Edge Impulse Studio. Select the option **Classification** on the app, and when it is running, start testing your images, confirming that the model is working with live and real data:



Lab

- To deploy the model on the Nicla, Edge Impulse Studio will package all the needed libraries, preprocessing functions, and trained models, downloading them to your computer. Select Deployment on the left side of the window.



Lab

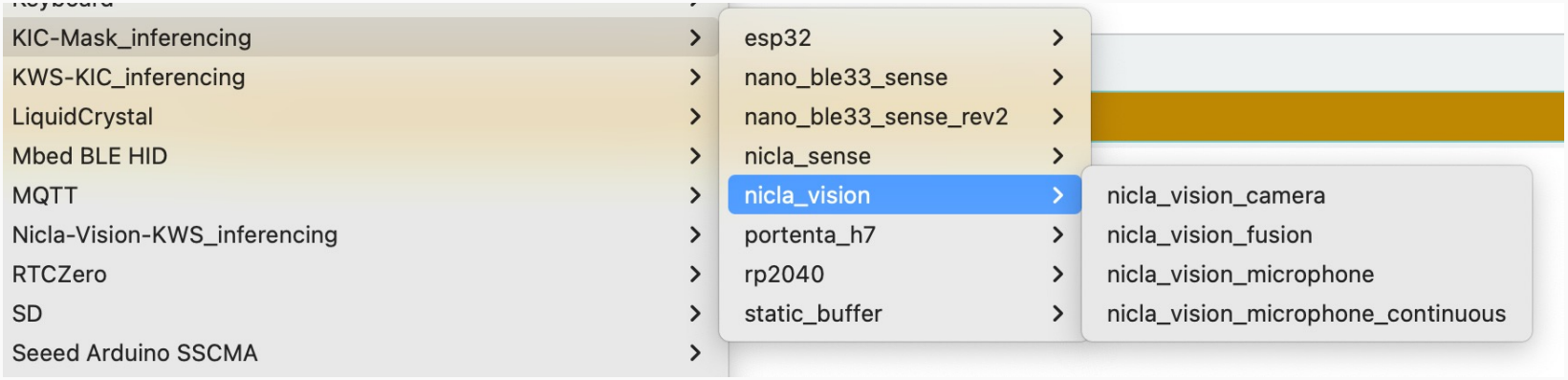
- You should select the option **Arduino Library** and at the bottom, select **Quantized (Int8),** de-select the option **Enable EON Compiler.** You should select the target device (lowest part of the screen). You can now press the button Build.

- When the Build button is pressed, a Zip file will be created and downloaded to your computer.

- On your Arduino IDE, go to the **Sketch** tab and select the option **Include Library → Add .ZIP Library**, and Choose the new .zip file you have downloaded.

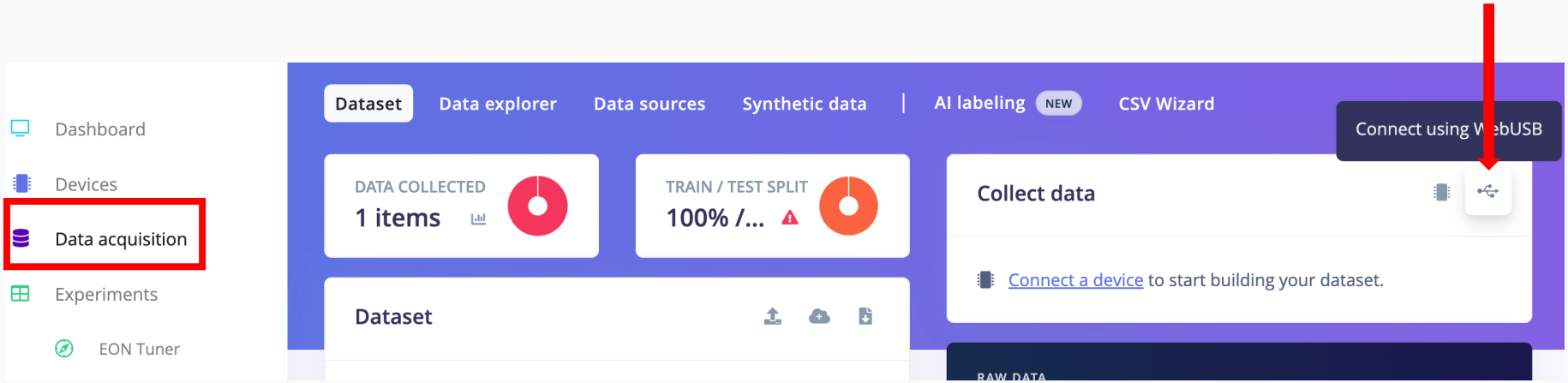
Lab

- In your Arduino IDE, go to the File →Examples tab and look for your project, and select Nicla Vision:



Lab

- If you want to collect images using the Nicla directly, follow these steps:



Make sure you are using Chrome as WebUSB is only available in Chrome!

Lab: install the Edge Impulse firmware first

- The Nicla Vision does **not ship with the Edge Impulse firmware**: flash it once before collecting images
- The official firmware supports **all onboard sensors, including the camera**; the ingestion sketches do not support the camera
- Then connect from **Data acquisition** via WebUSB in Chrome

1. Connect the board
Micro-USB cable; if flashing fails, press RESET twice for the bootloader



2. Download the firmware
Latest Edge Impulse firmware for Nicla Vision; unzip it

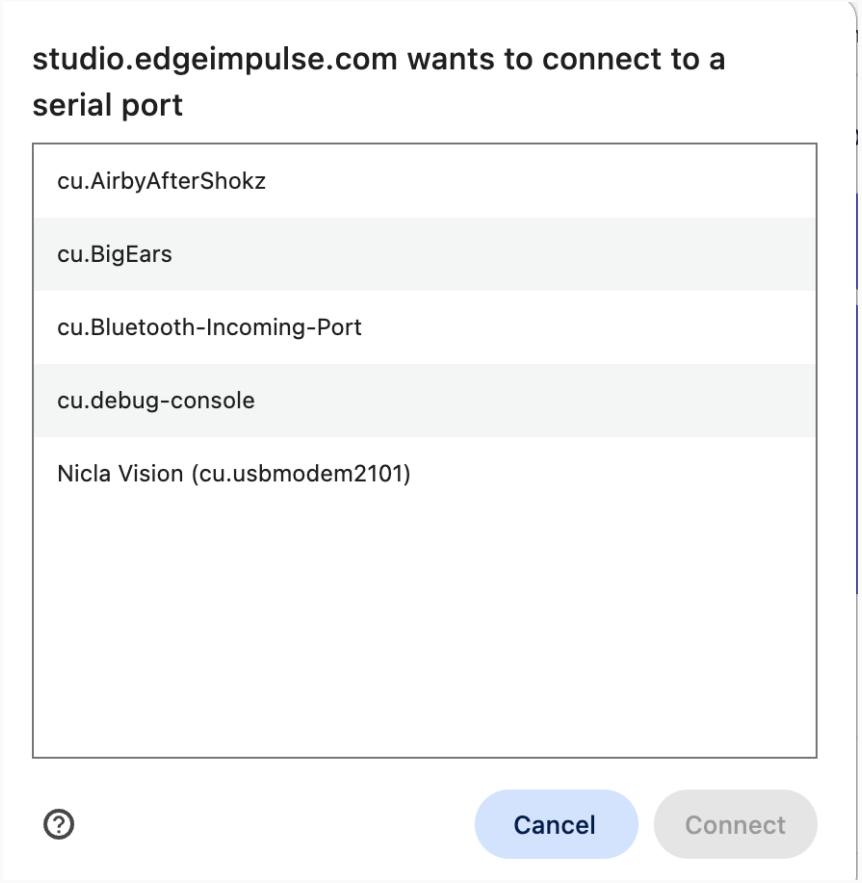


3. Run the flash script
flash_windows.bat, flash_mac.command or flash_linux.sh; then press RESET once

Firmware: cdn.edgeimpulse.com/firmware/arduino-nicla-vision.zip

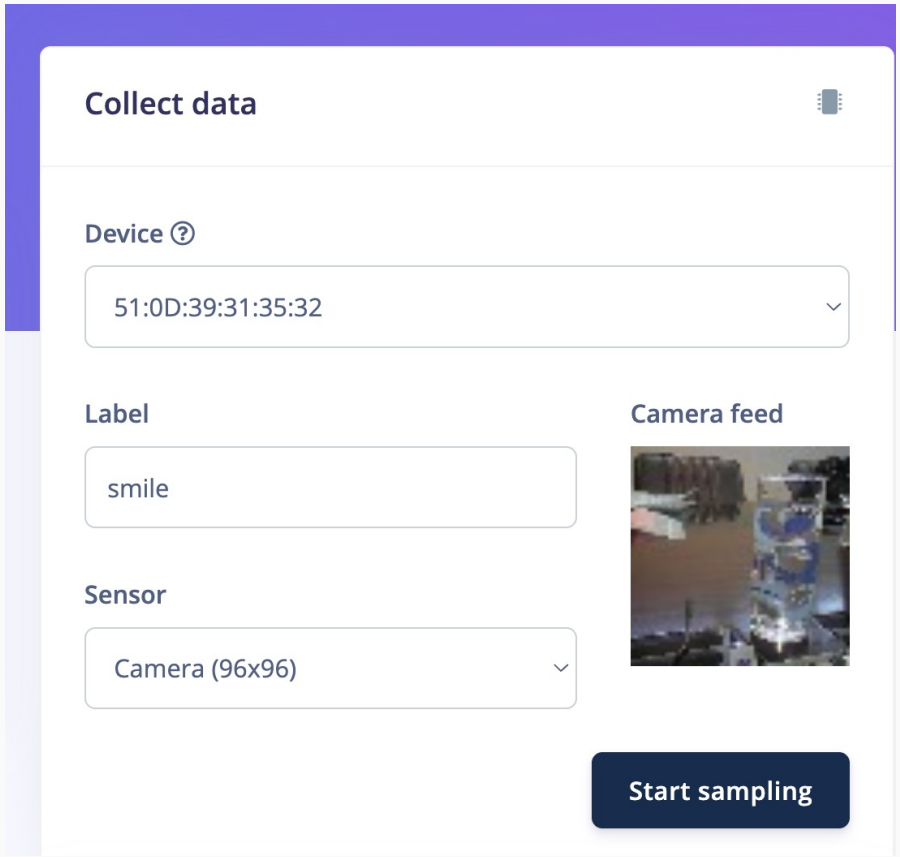
Lab

- Select the USB port where the Nicla is connected:



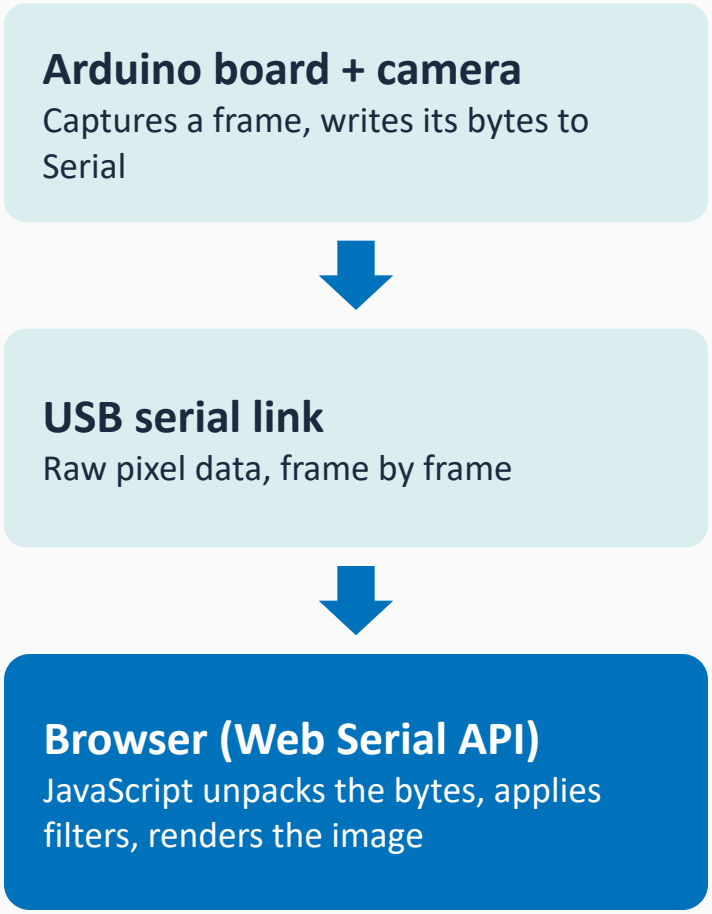
Lab

- On the right you will see the live camera feed and you will be able to assign a label to the images you collect. Click on Start sampling to save an image.



Lab: Web Serial Camera

- Stream the board's camera **straight to a web browser** over USB, with no extra software
- The sketch sends the **raw image bytes** over Serial; JavaScript in the page unpacks and draws each frame
- The web app's source includes **example image filters** that edit pixel data
- Sketch: **CameraCaptureWebSerial** (IDE: Examples → Camera)
- Try it: arduino.github.io/labs-pages/web-serial-camera



Key takeaways

1 Computer vision is hard

To a computer, an image is only a grid of pixel values (0–255); meaning has to be extracted.

2 Convolutions find features

A small filter slid across the image responds to big changes in value, such as edges and lane lines.

3 CNNs learn their own features

Networks like AlexNet learn the filters, then combine them into higher-level features to classify objects.

4 Microcontrollers need tiny models

MobileNet v1 needs 64× the 1 MB RAM of our board: use 96×96 grayscale input, a small α and int8.

5 Small datasets can still work

Balanced classes, data augmentation and transfer learning reached ~97% test accuracy in the lab.

6 Edge Impulse runs the workflow

Collect, train, convert and deploy to the Nicla Vision as an Arduino library, collecting data on the board itself.

Thanks!

TinyML whispers low,
AI in palms, humble glow,
Minds bloom, small yet grow.

mzennaro@ictp.it